

Where to Build Affordable Housing?

Evaluating the Trade-offs of Location

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Any opinions and conclusions expressed herein are those of the authors and do not reflect the views of the U.S. Census Bureau. The Census Bureau has reviewed this data product to ensure appropriate access, use, and disclosure avoidance projection of the confidential source data (Project #7529092, CBDRB-FY2023-CES010-017, CBDRB-FY2023-CES023-006, CBDRB-FY24-CES014-012, CBDRB-FY25-CES025-007, CBDRB-FY26-0197).

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 - Concerns over segregation, concentration of poverty, ...
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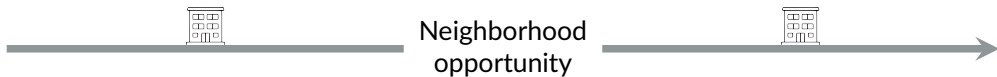
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Externalities / other: effects on segregation, neighbors, children, ...

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Neighborhood
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School quality (HUD)	4th ptile
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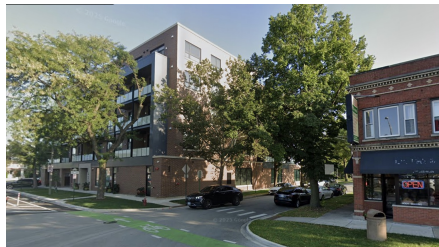


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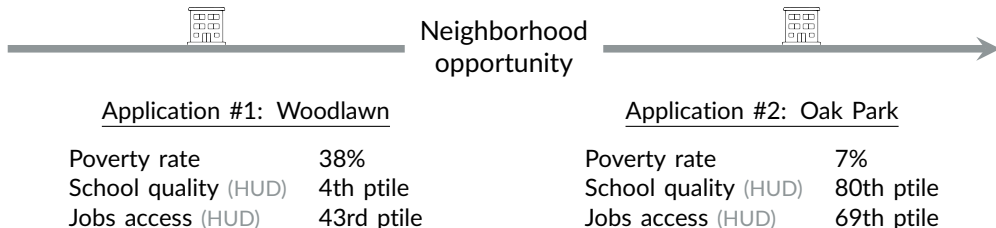
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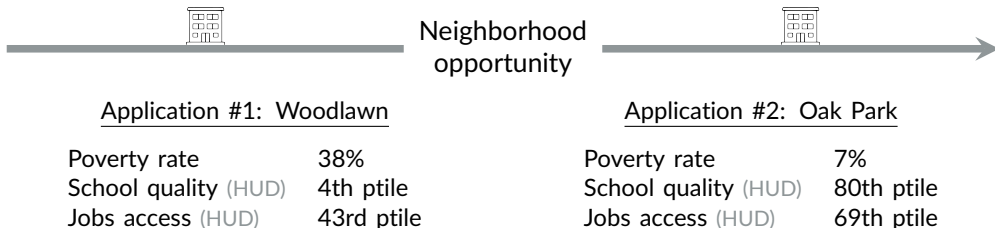
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Typical Q4 development

+ 16%
+ 94%
- 11%
- 37%

This paper

Setting: Low-Income Housing Tax Credit (LIHTC)

- Largest and fastest-growing AH program (~3 million units)

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Data: Administrative records linked both LIHTC and market-rate units

- Tax returns, residential addresses, and demographics for universe of US residents

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Descriptive evidence: Who lives in affordable housing?

1. Among those eligible, who lives in LIHTC *on average*?
2. How does this vary *across neighborhoods*?

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Descriptive evidence: Who lives in affordable housing?

Structural model: Residential choice with market-rate and (rationed) AH options

1. Identify the mechanisms behind differences across neighborhoods
 ~> Role of preference heterogeneity vs. rationing process
2. Quantify value to households, distribution of assistance, effects on segregation, ...
3. Compare effects of location to other policy levers

Preview of results

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- (+) Reduces racial/ethnic and economic segregation
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Current policy levers available post-construction have relatively limited effects

- E.g., lower income limits, income-based rents, fair lotteries, ...

Setting, data, and working with the Census

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3.4 million units built since 1987

- About a fourth of all multi-family construction

► LIHTC supply-side



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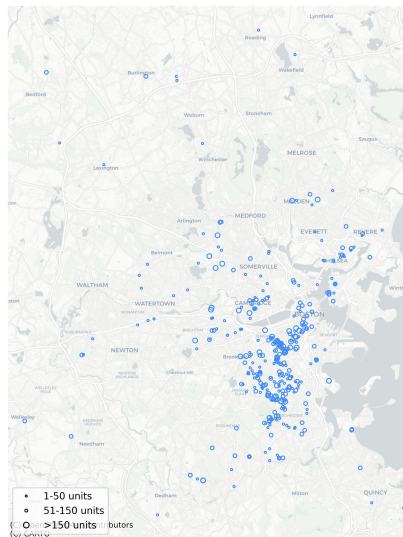
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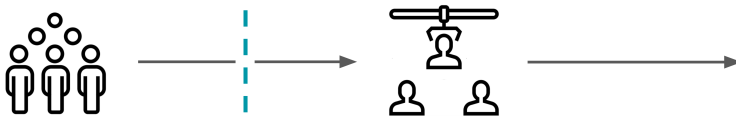
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Eligibility

Means-testing based on city-level income limits (adjusted for household size)

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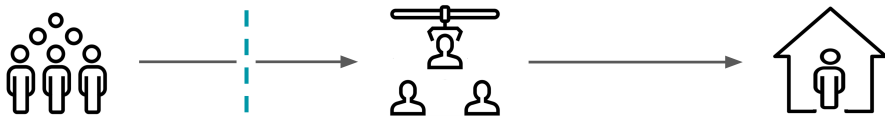
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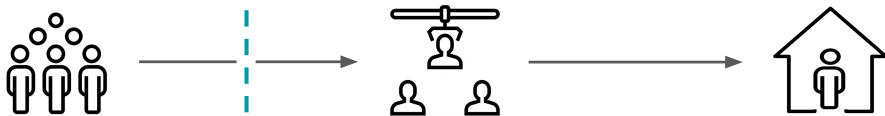
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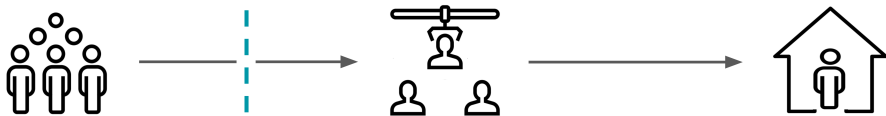
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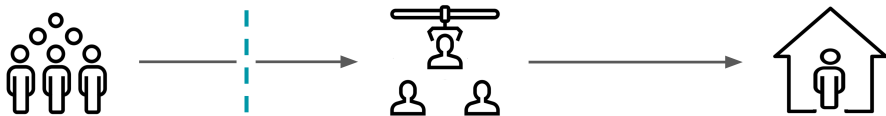
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Data for this project

Only possible thanks to collaborations between Census, IRS, and HUD

- **HUD:** locations and characteristics of LIHTC units
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Allows for extremely rich view into both market-rate and LIHTC households

- Annual income, sources of income, education, household structure, ...
- “Childhood family income” if claimed as dependent post-1994
- Where households lived before/after
- **Main sample:** renter households living in 50 most populous MSAs, 2010-2019
 - Observe both eligible households and those that receive a unit
 - Do *not* observe applications for units

Some ongoing projects using this data infrastructure

Subsidized Housing in the US: Household Dynamics around Move-in

with Rebecca Diamond, Winnie van Dijk, and John Voorheis

- **Goal:** Estimate treatment effects of receiving Public Housing, LIHTC, and Housing Choice Vouchers on household outcomes
 - Labor supply, household structure, household/neighborhood characteristics, ...
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Long-run Effects of Place-Based Policies: Evidence from Empowerment Zones

with Ari Binder, Eric Chyn, and Rob Collinson

- **Goal:** Estimate long-run treatment effects of the 1990s-era Empowerment Zone program
 - Home prices/rents, income of residents, displacement, long-run income of children, ...
- Evaluate effects on incumbent residents, new residents, firms, neighboring areas, ...
- Were the effects persistent after the subsidies expired?

Projects involving partnerships with local agencies

[Waldinger \(2021\)](#): Partnership with Cambridge Housing Authority to study their public housing allocation processes

- **Key takeaway:** Offering applicants more choice in where they live creates a trade-off between providing good matches and targeting assistance to the highest-need households

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[Bergman et al. \(2023\)](#): Experiment with Seattle to help voucher-recipients use their vouchers in higher-opportunity neighborhoods

- **Key takeaway:** Can create durable “moves to opportunity” by providing information, assistance navigating housing search, dealing with landlords, etc.

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[Collinson and Gibbs \(2025\)](#): Partnership with Miami HA to use their past lotteries to estimate treatment effects on kids

- **Key takeaway:** Vouchers improve children’s early literacy skills, school readiness, and test scores. Public housing has no detectable effects.

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5. Results go through disclosure review before they leave the Census
 - E.g., check minimum observation counts for all statistics

Who lives in LIHTC housing?

Who lives in LIHTC developments?

- 1) Among eligible, who receives a LIHTC unit **on average?**
- 2) How does it vary **across neighborhoods?**

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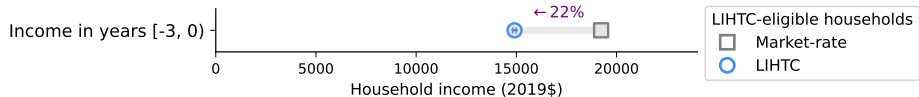
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- Among LIHTC-eligible renters, regress

Household characteristic $\sim$ isLIHTC + MSA $\times$ year

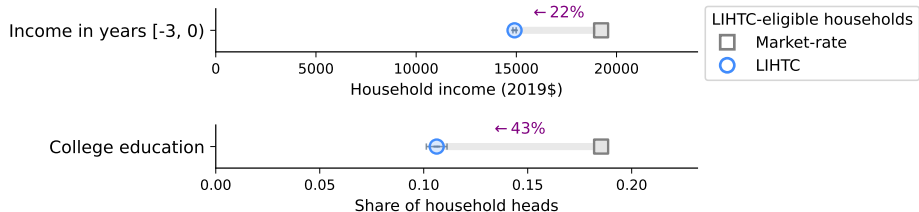
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Fact 1a: Households in LIHTC exhibit greater need



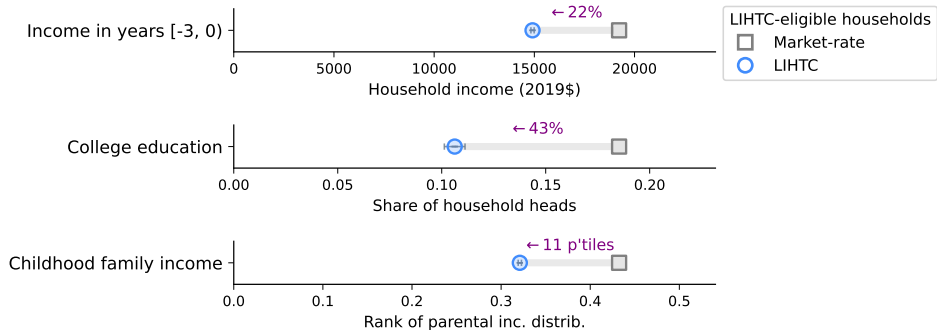
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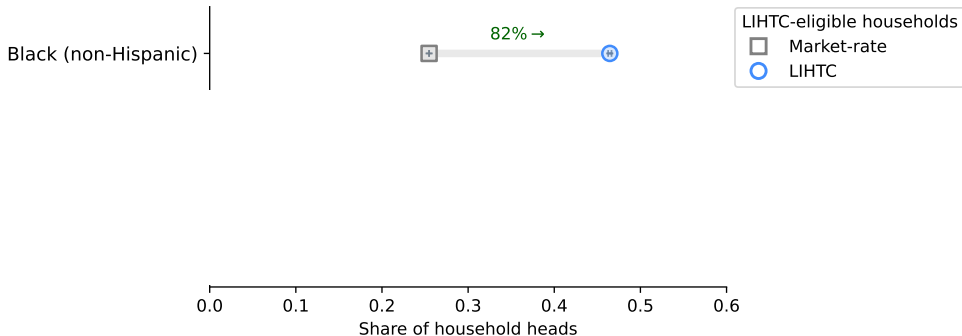
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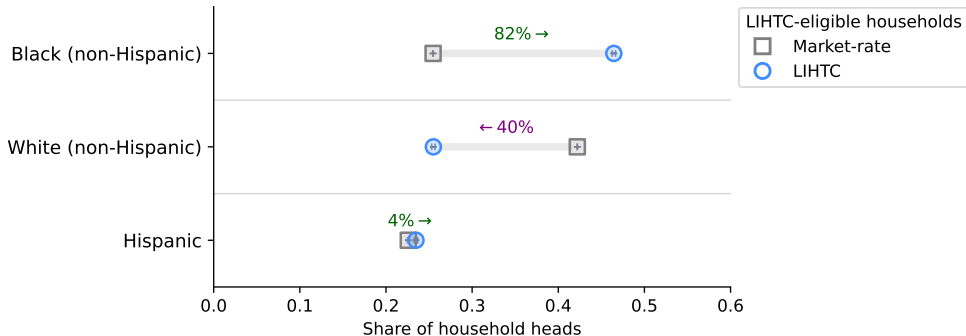
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Fact 1b: LIHTC household heads more likely to be Black



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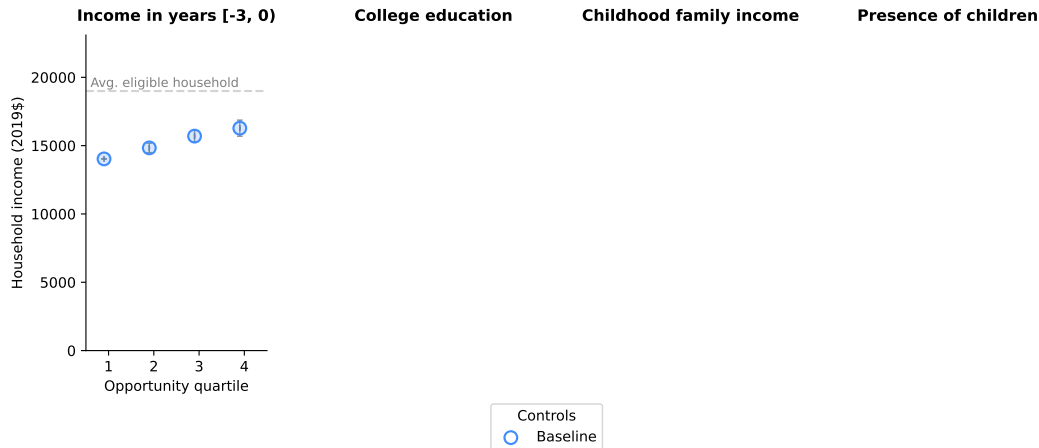
2) How does it vary **across neighborhoods**?

- Within sample of LIHTC renters, regress

$$\text{Household characteristic} \sim \text{opportunityQuartile} + \text{MSA} \times \text{year}$$

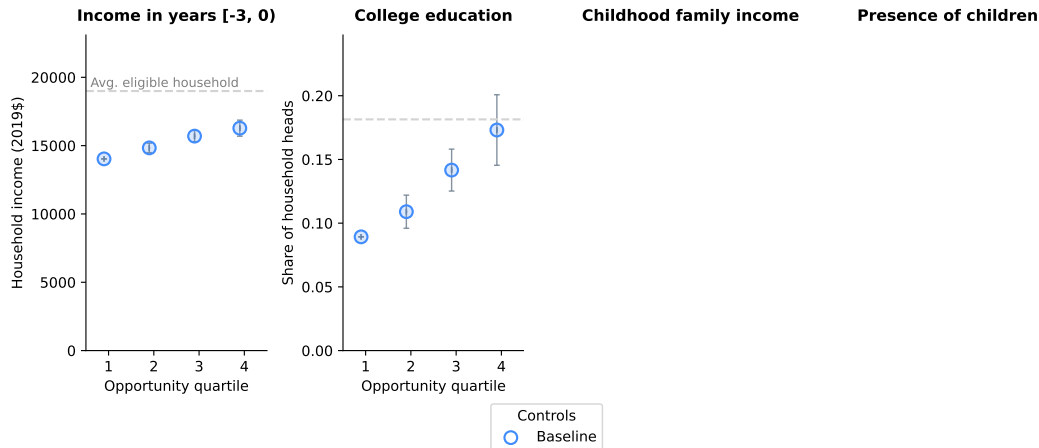
- LIHTC rent and income limits are **fixed** within a city

Fact 2a: Households in high-opportunity nbhds exhibit lower need



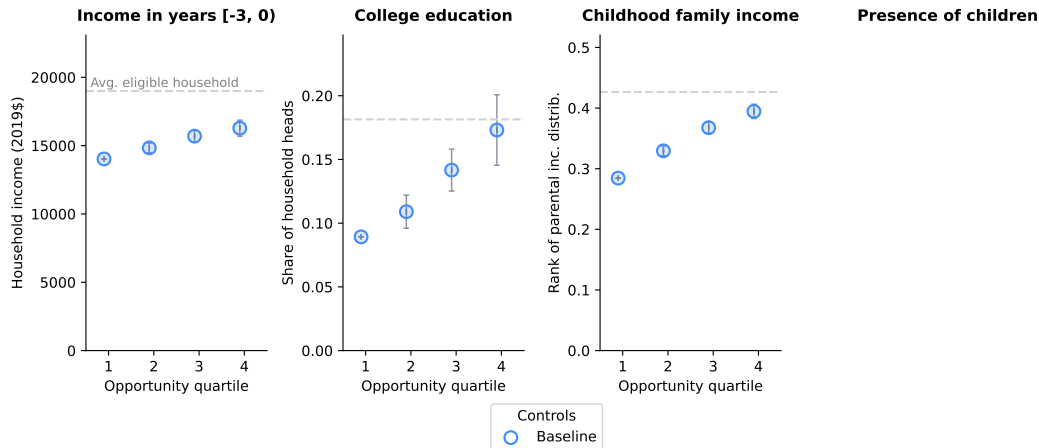
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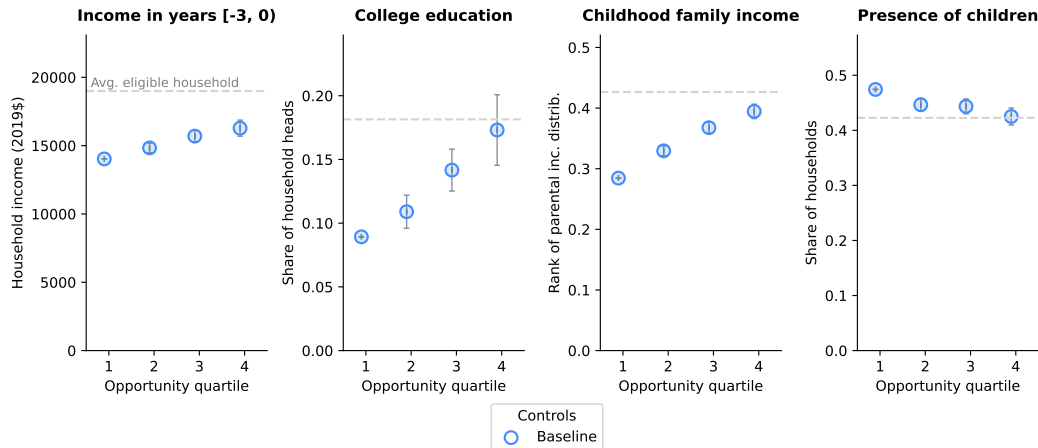
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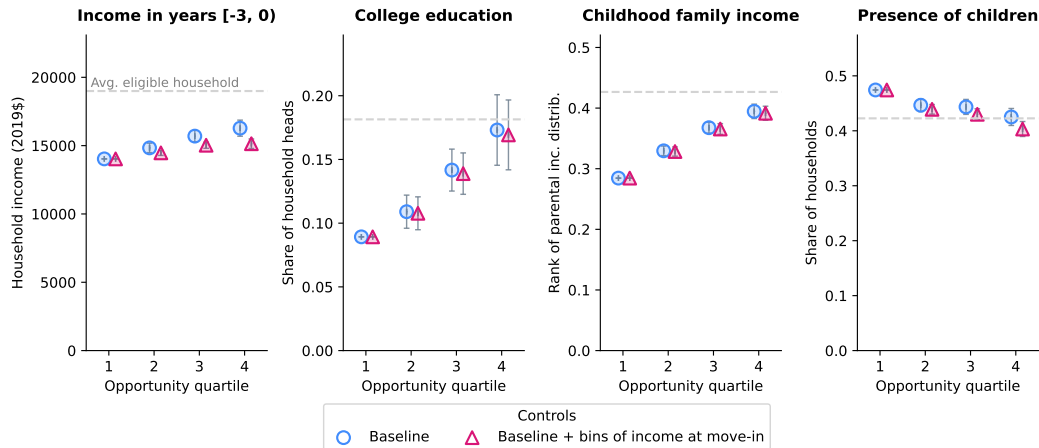
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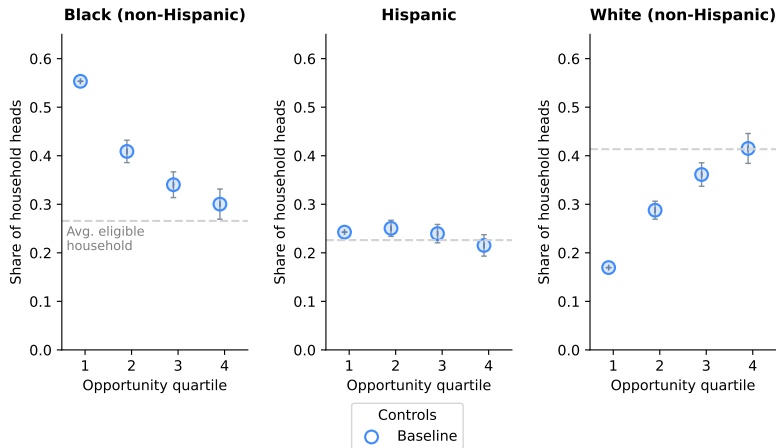
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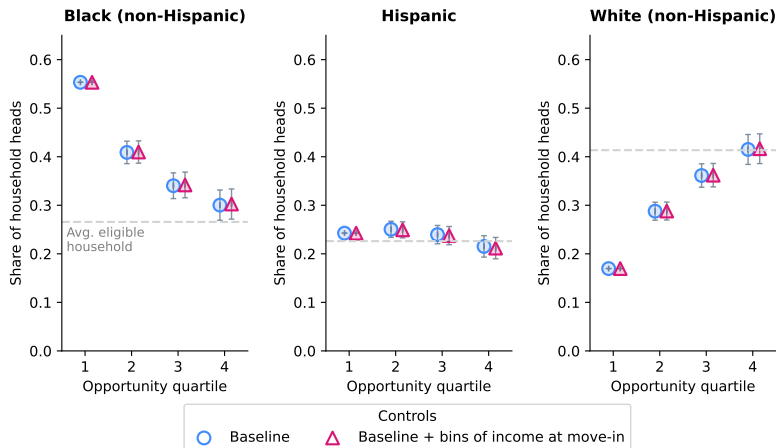
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Fact 2b: Share Black/Hispanic decreasing in nbhd opportunity



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2) How does it vary **across neighborhoods?**

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Goals of structural model

- Explain descriptive patterns

 - ↪ changes in *who applies* or changes in *which applicants are allocated unit?*

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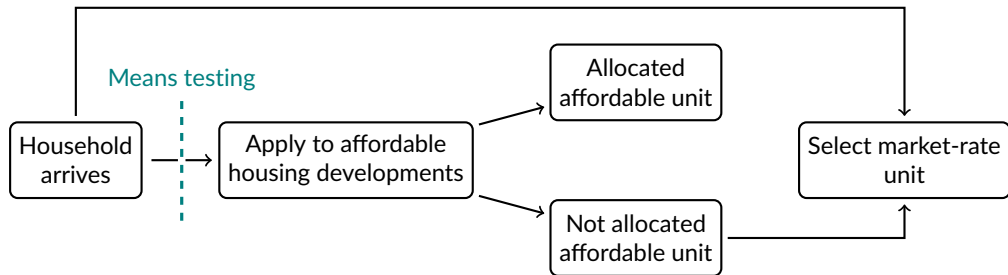
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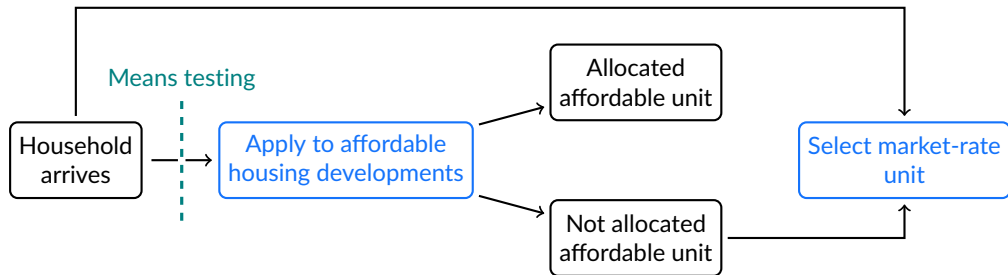
- Explain descriptive patterns
 - ↪ changes in *who applies* or changes in *which applicants are allocated unit?*
- Quantify tradeoffs using the **model**, **cost estimates**, and outside estimates
 - **Value:** value to households and costs to provide
 - **Distribution:** who benefits from assistance
 - **Externalities / other:** effects on segregation, children, neighbors

Model with affordable & market-rate housing

Static residential choice with affordable housing options



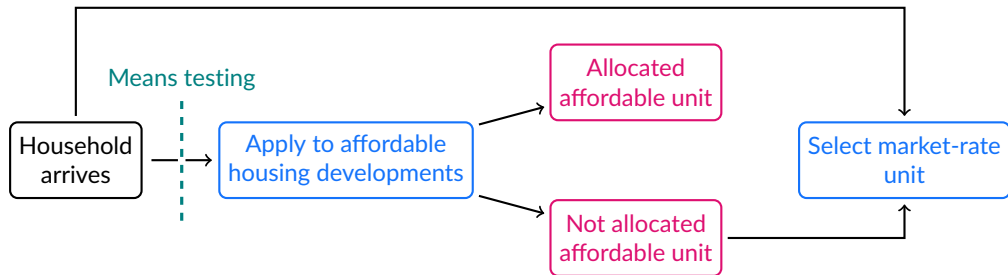
Static residential choice with affordable housing options



Two key ingredients:

1. Preferences for housing and neighborhoods

Static residential choice with affordable housing options



Two key ingredients:

1. Preferences for housing and neighborhoods
2. Rationing process for affordable housing

The model in one slide

Household preferences

- Preferences over housing & nbhd chars. that vary by household observables
- May value AH more/less than observably equivalent market-rate unit

Developers ration using 'weighted lottery'

- Up/down weight an applicant household based on characteristics (e.g., income)
- Offer probabilities clear the market (more applicants $\implies$ lower offer rates)

Estimation for Chicago MSA

1. Market-rate choices to learn preferences on observables (Bayer and Timmins, 2007)
2. Moves in/out of AH to separate developer weights from preferences specific to AH

Summary of key parameter estimates

1) Heterogeneity in preferences for housing/nbhd chars.

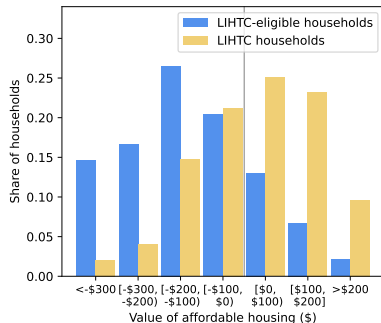
- Similar to existing estimates (Bayer, Ferreira and McMillan, 2007; Calder-Wang, 2021)
 - E.g., Higher income → less sensitive to rents, more weight on amenities like school quality

Summary of key parameter estimates

1) Heterogeneity in preferences for housing/nbhd chars.

2) Heterogeneity in preferences for affordable housing specifically

- Avg eligible household values unit in median opportunity nbhd at **-\$67/mo** compared to similar market-rate unit
 - Gap is smaller in higher-opportunity nbhds
 - Substantial heterogeneity. \$13 for Black/Hispanic-led households, -\$172 for White/other
 - Value of AH increasing in measures of need (↪self-targeting)

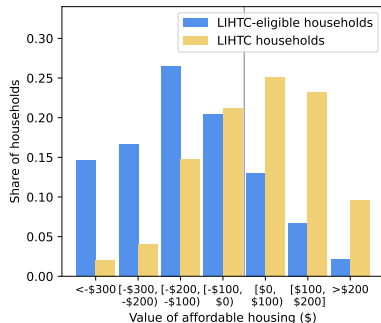


Summary of key parameter estimates

1) Heterogeneity in preferences for housing/nbhd chars.

2) Heterogeneity in preferences for affordable housing specifically

- Avg eligible household values unit in median opportunity nbhd at **-\$67/mo** compared to similar market-rate unit
 - Gap is smaller in higher-opportunity nbhds
 - Substantial heterogeneity. \$13 for Black/Hispanic-led households, -\$172 for White/other
 - Value of AH increasing in measures of need (↪self-targeting)
- Many place *positive* value on affordable housing
 - [Reid \(2018\)](#): In surveys, tenants highlight reliable landlords, rent stability, sense of community, “safe haven”



Summary of key parameter estimates

- 1) Heterogeneity in preferences for housing/nbhd chars.
- 2) Heterogeneity in preferences for affordable housing specifically
- 3) Screening favors high-need and Black/Hispanic, but less so in higher-opportunity nbhds 
 - Applicants that are high-need or Black/Hispanic are 5-25% more likely chosen than the average applicant
 - Small differences across neighborhoods – gaps in offer rates generally decreasing in nbhd opportunity

Decomposing the role of preferences vs screening

Population	Share of households (existing LIHTC developments)	
	All renters	LIHTC tenants
Black/Hispanic, high-need	0.18	0.53
Black/Hispanic, moderate-need	0.29	0.30
White/other, high-need	0.08	0.05
White/other, moderate-need	0.45	0.11

Note: This table documents the simulated share of households in existing LIHTC developments under different scenarios. $\phi = 0$ shuts down screening ('fair lottery') and $\alpha = 0$ shuts down any preferences specific to affordable housing.

Define need based on prediction of future income using observables at move-in

- 'High-need' if in bottom quartile of renters, adjusted for household size and age



Decomposing the role of preferences vs screening

Population	Share of households (existing LIHTC developments)		
	All renters	LIHTC- eligible	LIHTC tenants
Black/Hispanic, high-need	0.18	0.33	0.53
Black/Hispanic, moderate-need	0.29	0.25	0.30
White/other, high-need	0.08	0.14	0.05
White/other, moderate-need	0.45	0.28	0.11

Note: This table documents the simulated share of households in existing LIHTC developments under different scenarios. $\phi = 0$ shuts down screening ('fair lottery') and $\alpha = 0$ shuts down any preferences specific to affordable housing.

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Decomposing the role of preferences vs screening

Population	Share of households (existing LIHTC developments)			
	All renters	LIHTC- eligible	$\alpha = 0$, $\phi = 0$	LIHTC tenants
Black/Hispanic, high-need	0.18	0.33	0.40	0.53
Black/Hispanic, moderate-need	0.29	0.25	0.31	0.30
White/other, high-need	0.08	0.14	0.08	0.05
White/other, moderate-need	0.45	0.28	0.20	0.11

Note: This table documents the simulated share of households in existing LIHTC developments under different scenarios. $\phi = 0$ shuts down screening ('fair lottery') and $\alpha = 0$ shuts down any preferences specific to affordable housing.

Define need based on prediction of future income using observables at move-in

- 'High-need' if in bottom quartile of renters, adjusted for household size and age

$\alpha = 0$ removes any preferences specific to affordable housing

$\phi = 0$ uses a fair lottery to allocate

Decomposing the role of preferences vs screening

Population	Share of households (existing LIHTC developments)				
	All renters	LIHTC- eligible	$\alpha = 0$, $\phi = 0$	$\phi = 0$	LIHTC tenants
Black/Hispanic, high-need	0.18	0.33	0.40	0.47	0.53
Black/Hispanic, moderate-need	0.29	0.25	0.31	0.32	0.30
White/other, high-need	0.08	0.14	0.08	0.05	0.05
White/other, moderate-need	0.45	0.28	0.20	0.16	0.11

Note: This table documents the simulated share of households in existing LIHTC developments under different scenarios. $\phi = 0$ shuts down screening ('fair lottery') and $\alpha = 0$ shuts down any preferences specific to affordable housing.

Define need based on prediction of future income using observables at move-in

- 'High-need' if in bottom quartile of renters, adjusted for household size and age

$\alpha = 0$ removes any preferences specific to affordable housing

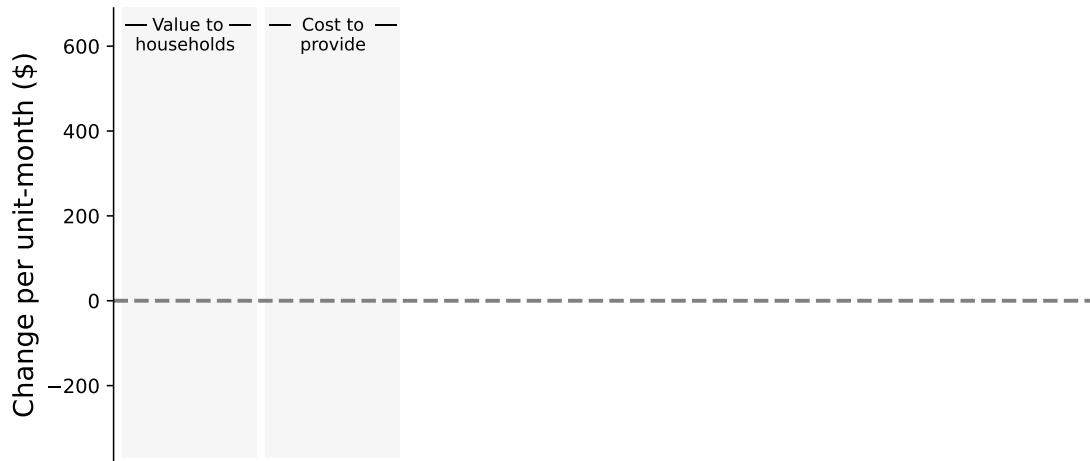
$\phi = 0$ uses a fair lottery to allocate

Where to build affordable housing?

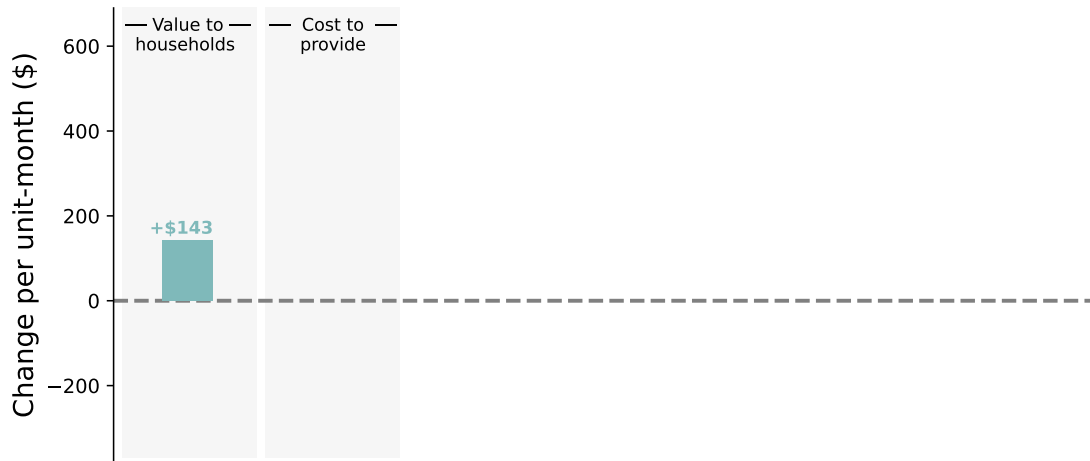
Main exercise: simulate adding additional AH units to housing stock

- Typical development: 100 units, avg. distribution of bedroom sizes [▶ LIHTC supply-side](#)
- Vary in which neighborhood it is placed, holding fixed other options

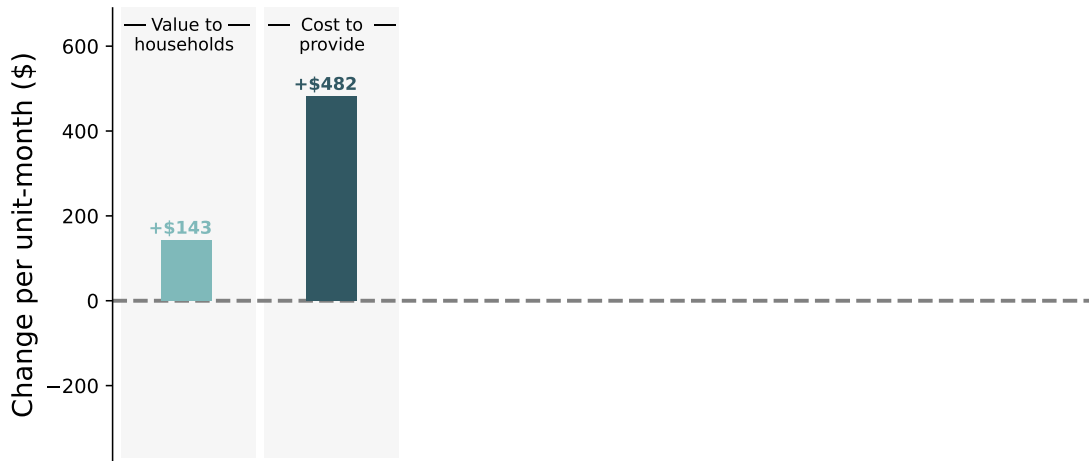
Effects of a new development in top vs. bottom quartile



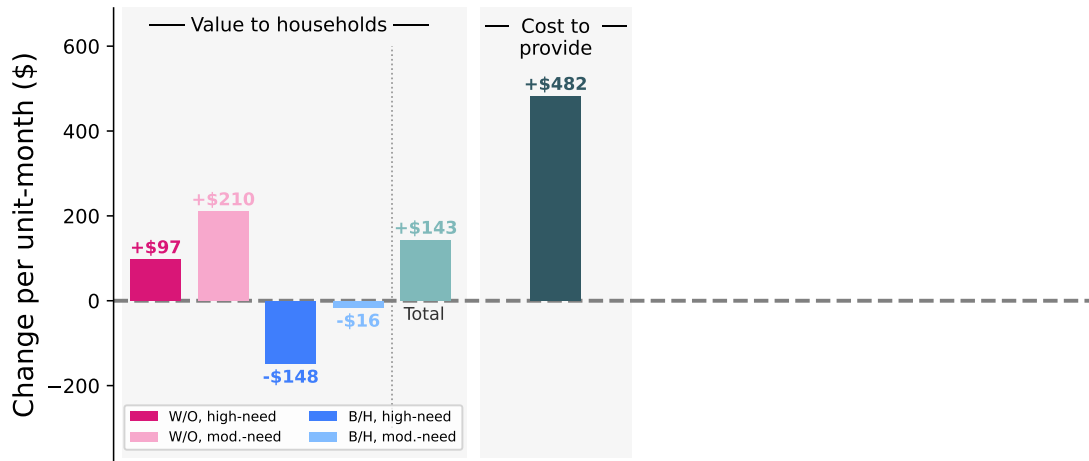
Effects of a new development in top vs. bottom quartile



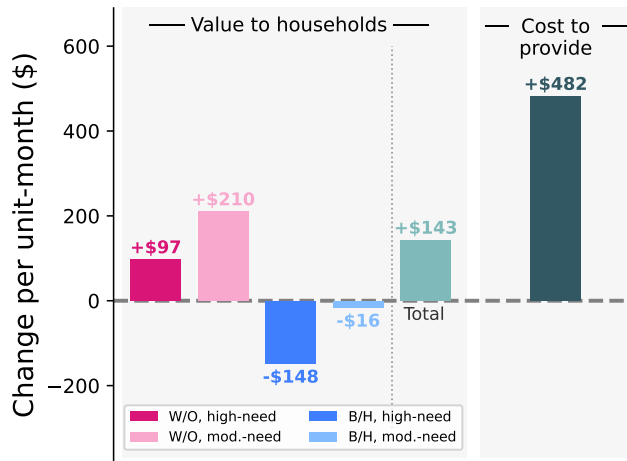
Effects of a new development in top vs. bottom quartile



Effects of a new development in top vs. bottom quartile



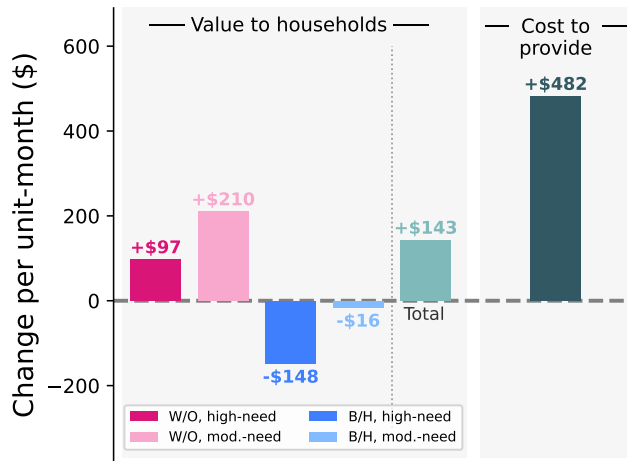
Effects of a new development in top vs. bottom quartile



— Other policy goals —

- (+) Racial/ethnic integration
- (+) Economic integration
- (+) Upward mobility of children
(Chetty et al., 2026)
- (-) Spillovers on neighbors
(Diamond and McQuade, 2019)

Effects of a new development in top vs. bottom quartile



— Other policy goals —

- (+) Racial/ethnic integration
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- (+) Upward mobility of children
(Chetty et al., 2026)
- (-) Spillovers on neighbors
(Diamond and McQuade, 2019)

Conclusion

This paper: Choice of location is implicitly a choice of tenants

- Building in higher-opportunity neighborhoods:
 - Provides more value for residents, improves long-run earnings of children, ...
 - But costs more and changes the distribution of assistance towards more moderate-need, predominantly white households

Conclusion

This paper: Choice of location is implicitly a choice of tenants

- Building in higher-opportunity neighborhoods:
 - Provides more value for residents, improves long-run earnings of children, ...
 - But costs more and changes the distribution of assistance towards more moderate-need, predominantly white households

Many opportunities for further research

Two questions I'm especially interested in:

1. How should government agencies manage the waitlists, lotteries, household priorities, eligibility testing, etc. when allocating assistance?
2. What are the trade-offs of providing assistance via vouchers, public housing, inclusionary zoning, and LIHTC?

Thank you!

cody.cook@yale.edu

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Affordable housing & moving to opportunity

Katz, Kling and Liebman (2001); Sinai and Waldfogel (2005); Kling, Liebman and Katz (2007); Turner, Popkin and Rawlings (2009); Baum-Snow and Marion (2009); Eriksen and Rosenthal (2010); Ellen, Horn and O'Regan (2016); Chetty, Hendren and Katz (2016); Chyn (2018); Ellen, Horn and Kuai (2018); Diamond and McQuade (2019); van Dijk (2019); Derenoncourt (2022); Almagro, Chyn and Stuart (2024); Bergman et al. (2023)

→ Choice of location is implicitly a choice of tenants, which affects distribution and segregation

Affordable housing & moving to opportunity

→ Choice of location is implicitly a choice of tenants, which affects distribution and segregation

Redistribution via in-kind transfers.

Nichols and Zeckhauser (1982); Blackorby and Donaldson (1988); Besley and Coate (1991); Currie and Gahvari (2008); Kleven and Kopczuk (2011); Mullainathan and Shafr (2013); Alatas et al. (2016); Finkelstein and Notowidigdo (2019); Lieber and Lockwood (2019); Deshpande and Li (2019)

→ 'Crowding out' in rationing process + potential distortions of private developers

Affordable housing & moving to opportunity

- Choice of location is implicitly a choice of tenants, which affects distribution and segregation

Redistribution via in-kind transfers.

- 'Crowding out' in rationing process + potential distortions of private developers

Methodology: residential choice + empirical market design

Residential choice: Bayer, Ferreira and McMillan (2007); Geyer and Sieg (2013); Galiani, Murphy and Pantano (2015); Diamond (2016); Bayer et al. (2016); Bergman, Chan and Kapor (2020); Calder-Wang (2021)

Empirical market design: Abdulkadiroğlu, Pathak and Roth (2005); Agarwal and Somaini (2020); Agarwal et al. (2021); Waldinger (2021); Huitfeldt, Marone and Waldinger (2024)

- Model with market-rate and *rationed* affordable housing options, without needing applications data

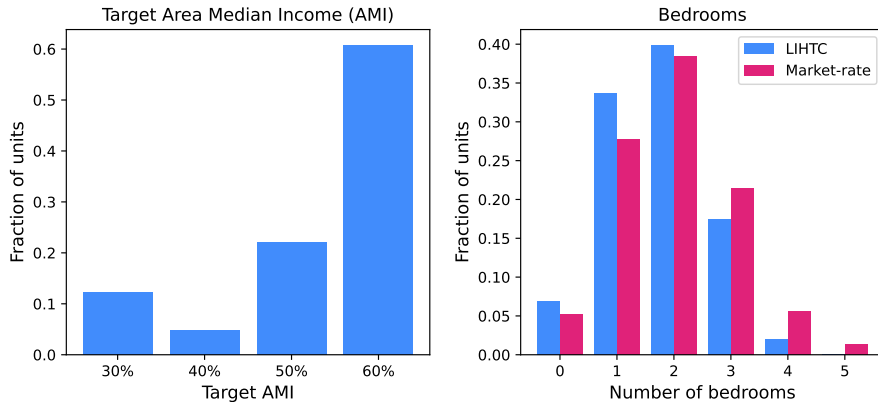
Building panel of LIHTC households

1. Link each individual to LIHTC address IDs (MAFIDs) to identify individuals in each unit
2. For each building, define a graph where individuals are nodes and edges are formed when they are married, claimer-dependent, or have shared other addresses.
3. Define households as all connected sets of the graph. Include claimed dependents not matched to building that year.
4. Define the household head as the highest wage earner (with age as tie breaker)

Building panel of market-rate households

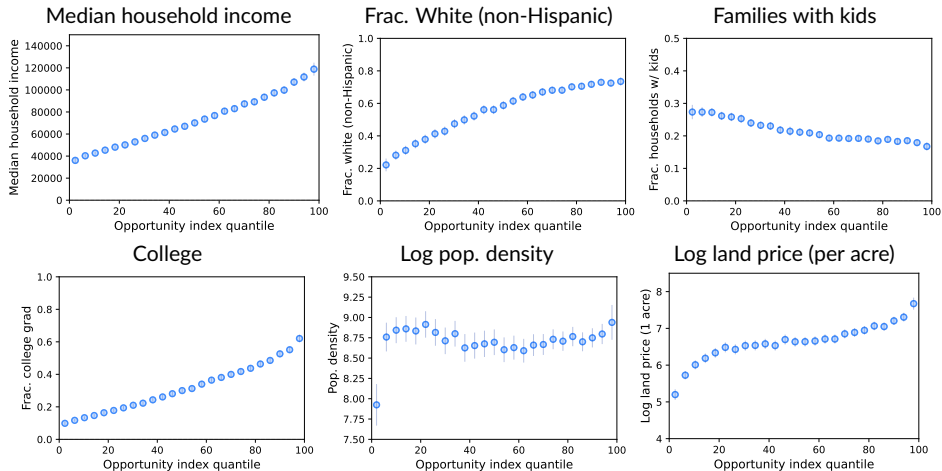
1. Subset ACS households to market-rate renters by excluding homeowners or renters in LIHTC, public housing, or project-based section 8
 - Note: that these excluded ACS renters are used to infer demand for affordable housing during estimation and counterfactuals
2. Merge on tax records, demographics, childhood family income, etc. to have equivalently defined characteristics as in LIHTC panel

Unit types in LIHTC and market-rate



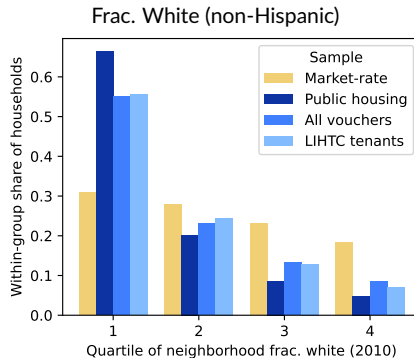
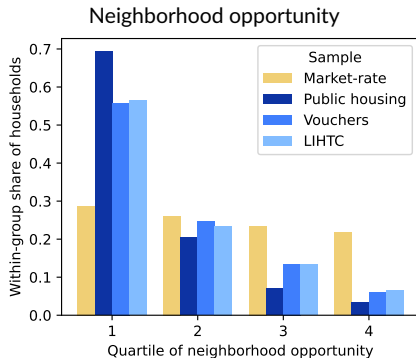
Note: Sample includes all LIHTC units in the 50 sample CBSAs that were built between 1995-2019 and do not target any special population. Market-rate units are from the ACS, 2010-2018.

Correlates of neighborhood opportunity



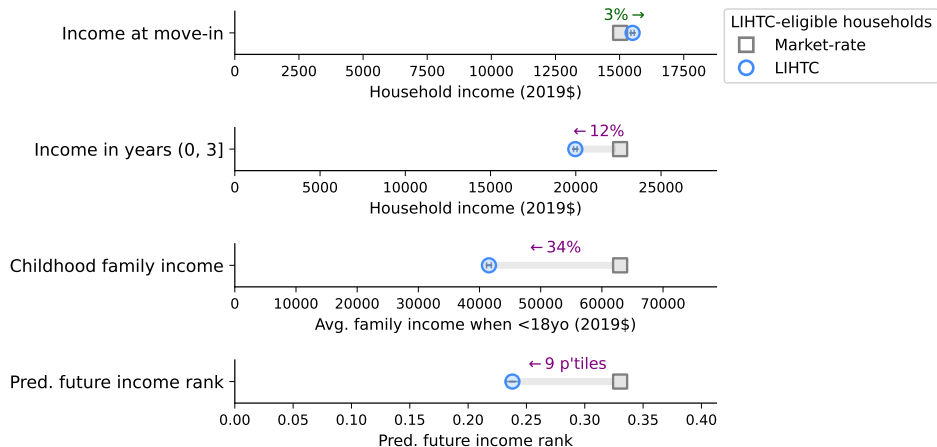
Note: Each figure documents the correlation between the within-CBSA percentile of our tract-level index of opportunity against different neighborhood characteristics for the 50 sample CBSAs. All characteristics except for land prices are sourced from the 2010 Census or 5-year ACS. Land values for each tract are estimated using data on historical land sales.

Affordable housing by neighborhood characteristics

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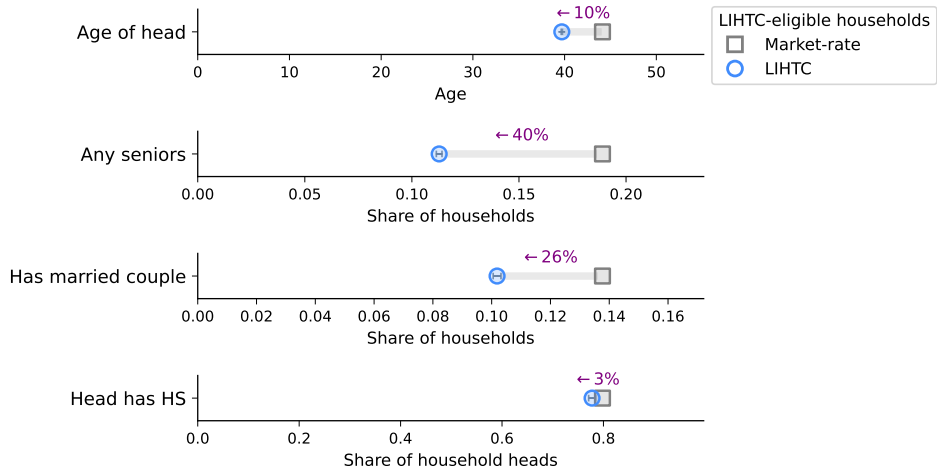
Note: Each bar is the fraction of households in the ACS in each within-MSA quartiles of neighborhood characteristics for the 50 sample CBSAs (2010-2018). Public housing and voucher data come from the HUD PICTRAC data.

Average differences: income

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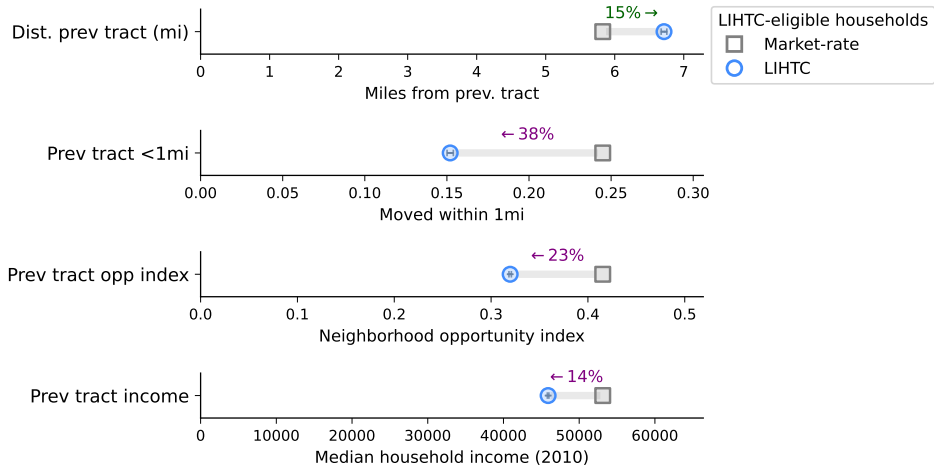
Note: Each point is based on coefficient from a regression of a household characteristics on an indicator for living in LIHTC with MSA $\times$ year fixed effects. Data include all households eligible for LIHTC. Outcomes are at the time surveyed (for market-rate) or time of move-in (for LIHTC). 95% confidence intervals are represented by grey lines.

Average differences: demographics

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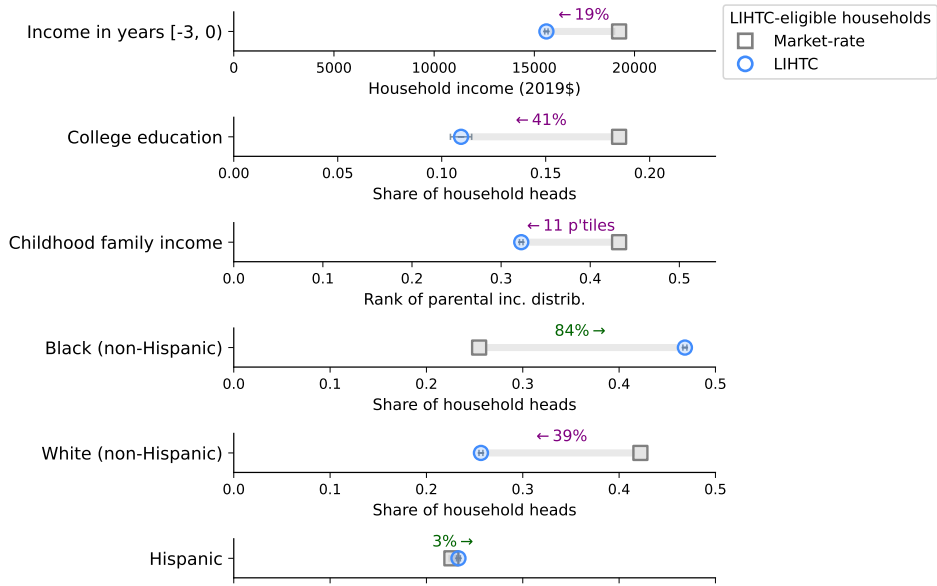
Note: Each point is based on coefficient from a regression of a household characteristics on an indicator for living in LIHTC with MSA $\times$ year fixed effects. Data include all households eligible for LIHTC. Outcomes are at the time surveyed (for market-rate) or time of move-in (for LIHTC). 95% confidence intervals are represented by grey lines.

Average differences: previous neighborhood characteristics

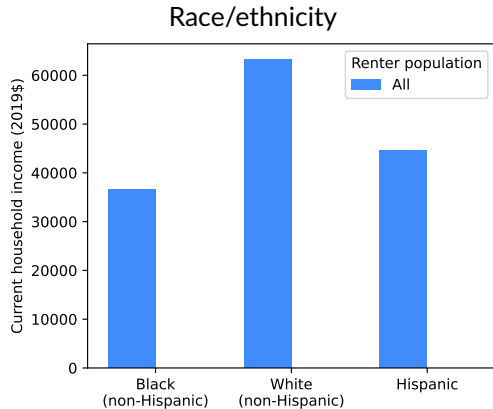
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Note: Each point is based on coefficient from a regression of a household characteristics on an indicator for living in LIHTC with MSA $\times$ year fixed effects. Data include all households eligible for LIHTC. Outcomes are at the time surveyed (for market-rate) or time of move-in (for LIHTC). 95% confidence intervals are represented by grey lines.

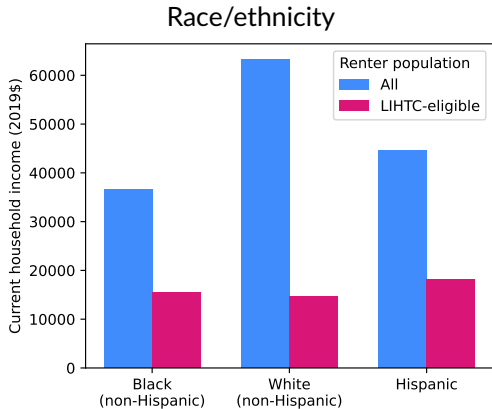
Average differences: using 50% AMI for eligibility

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Current income is a weak proxy for other characteristics

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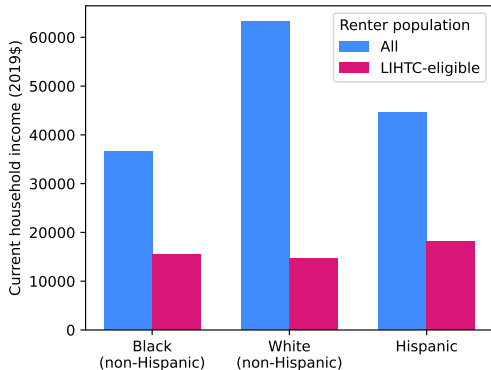
Current income is a weak proxy for other characteristics

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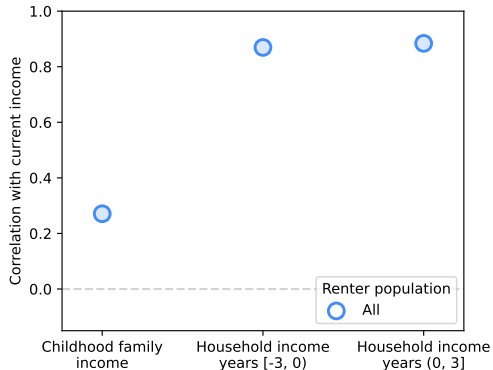
Current income is a weak proxy for other characteristics

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Race/ethnicity



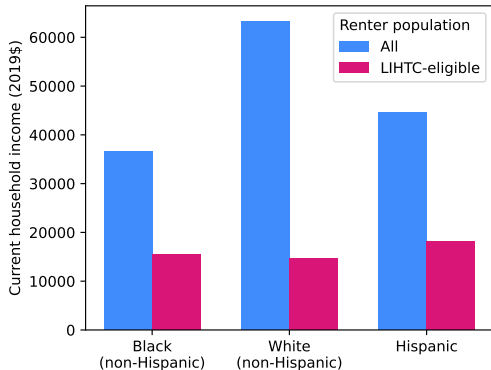
Childhood/past/future income



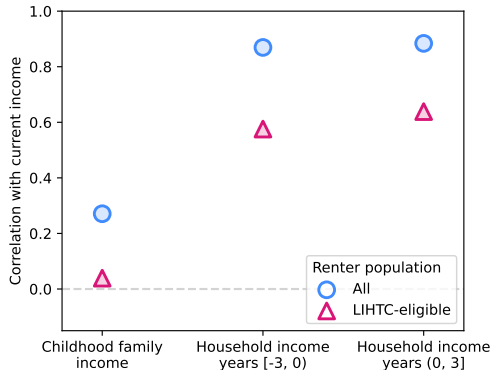
Current income is a weak proxy for other characteristics

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Race/ethnicity



Childhood/past/future income



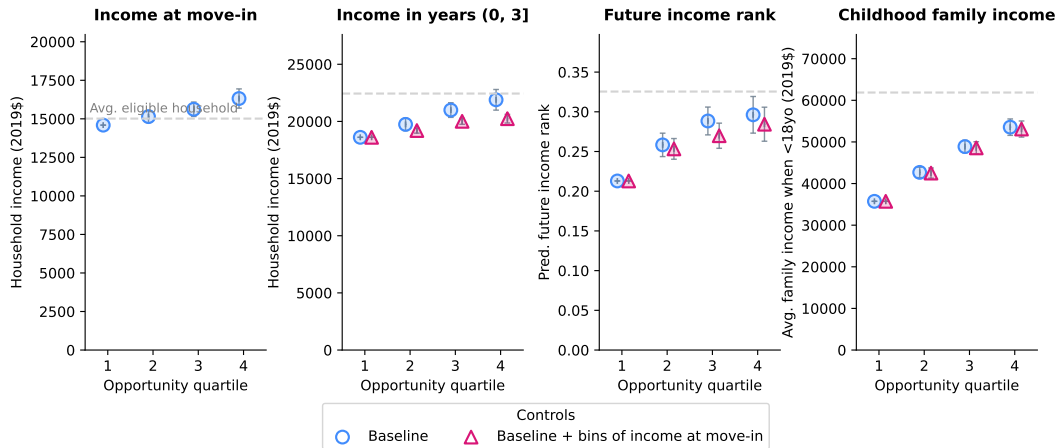
Current household income correlations

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Household characteristic	All renters	All LIHTC-eligible renters
Correlations: current AGI		
Avg. AGI in years [-3, 0)	0.8694	0.5756
Avg. AGI in years (0, 3]	0.8840	0.6385
Predicted future income rank	0.6760	0.3391
Childhood family income rank (household head)	0.2709	0.03903
Average current AGI by char.		
Black (non-Hispanic)	\$36,560	\$15,420
White (non-Hispanic)	\$63,290	\$14,640
Hispanic	\$44,680	\$18,070
Graduated college (household head)	\$89,660	\$18,120
No college degree (household head)	\$37,780	\$15,050
Graduated high school (household head)	\$58,970	\$16,850
No high school degree (household head)	\$22,810	\$11,180
Household has children (<18yo)	\$56,360	\$21,330
Household does not have children	\$51,750	\$11,420
Household has seniors (>64yo)	\$33,000	\$8,434
Household does not have seniors	\$57,620	\$17,600
Household has joint filers	\$92,180	\$28,680
Household has no joint filers	\$41,910	\$13,520

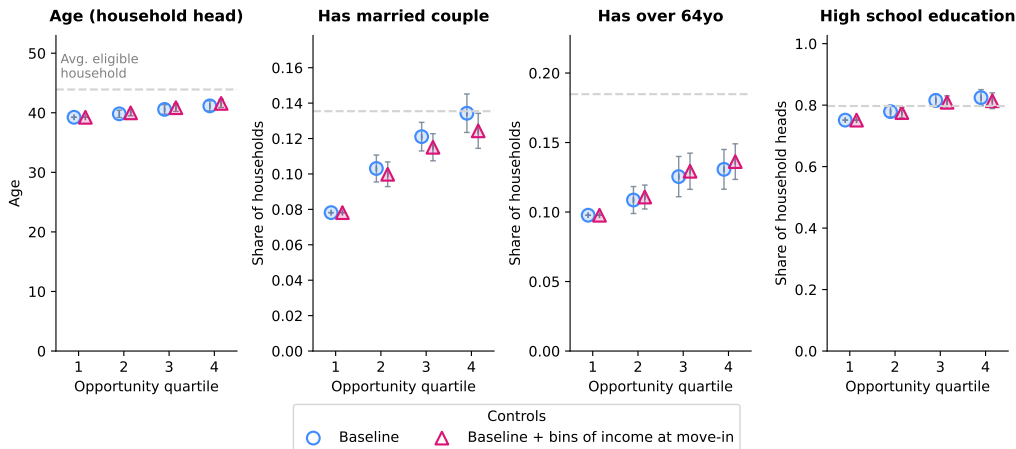
Note: This table documents the relationship between current Adjusted Gross Income (AGI) and other household characteristics for all renters and for all LIHTC-eligible renters in the ACS. The first three rows are correlations, while the remainder are the average AGI for the group indicated in the left column.

LIHTC households by neighborhoods: income

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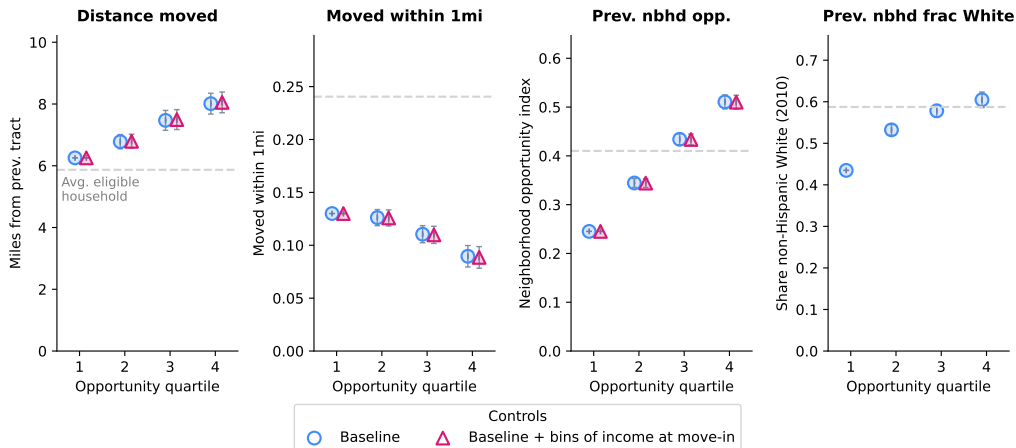
Note: Each point is based on coefficient from a regression of a household characteristics on indicators for neighborhood opportunity quartile. The baseline specification includes bedroom, income limit, and MSA $\times$ year fixed effects. Data include households at the time of move-in to LIHTC. 95% confidence intervals are represented by grey lines.

LIHTC households by neighborhoods: demographics

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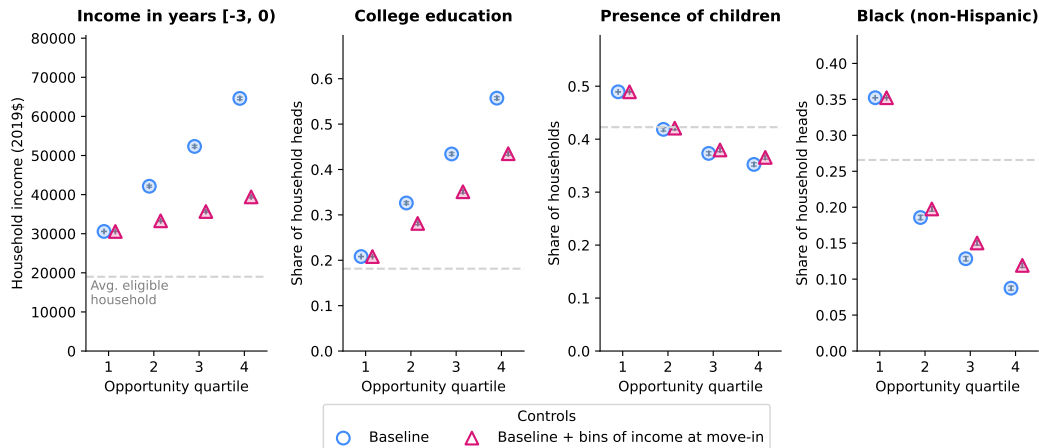
Note: Each point is based on coefficient from a regression of a household characteristics on indicators for neighborhood opportunity quartile. The baseline specification includes bedroom, income limit, and MSA $\times$ year fixed effects. Data include households at the time of move-in to LIHTC. 95% confidence intervals are represented by grey lines.

LIHTC households by neighborhoods: prev. nbhd

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Note: Each point is based on coefficient from a regression of a household characteristics on indicators for neighborhood opportunity quartile. The baseline specification includes bedroom, income limit, and MSA $\times$ year fixed effects. Data include households at the time of move-in to LIHTC. 95% confidence intervals are represented by grey lines.

Market-rate households by neighborhoods

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Note: Each point is based on coefficient from a regression of a household characteristics on indicators for neighborhood opportunity quartile. The baseline specification includes bedroom, income limit, and MSA $\times$ year fixed effects. Sample is ACS renters living in market-rate units. 95% confidence intervals are represented by grey lines.

1. Predict average income in 3-years after ACS survey

- **Method:** XGBoost (Chen and Guestrin, 2016)
- **Variables:** average household income and wages in 3 prior years, household income and wages in current year, indicator for any income, number of W2s, race/ethnicity of household head, number of household members, whether any children, whether any seniors, head of household age (and age squared), whether any joint filers, and characteristics of previous tract (median income, fraction White, and neighborhood opportunity index)
- R^2 on holdout sample of 0.861

2. Adjust using equivalence scale from Census

$$ES = \begin{cases} N_{\text{adults}}^{0.5} & \text{One and two adults} \\ (N_{\text{adults}} + 0.8 + 0.5(1 - N_{\text{children}}))^{0.7} & \text{Single parents} \\ (N_{\text{adults}} + 0.5N_{\text{children}})^{0.7} & \text{Other families} \end{cases}$$

3. Rank within distribution of renter households for 5-year age bins

- Age bin based on head of household. Renter distribution is for all 50 sample MSAs.

- **Housing:** indicators for bedrooms, building type (single-family, small apartment, big apartment), and whether <25 years old
- **Neighborhood:** race/ethnicity shares (2010 Census); share college-educated, log median household income, and log population density (2010 Census); HUD indices of school quality, jobs access, transit access, and poverty
- **Households:** race/ethnicity, presence of children, presence of seniors, presence of married couple, indicators for household size, whether has voucher, log current income, log average income in past three years, whether >0 current income, whether > 0 average income in past three years

For a candidate vector of parameters $\tilde{\theta}^{\text{MR}} = \{\tilde{\delta}, \tilde{\gamma}_\ell, \tilde{\beta}_\ell, \tilde{\kappa}\}$, the pseudo log-likelihood is:

$$\ell = \sum_t \sum_{i \in \mathcal{I}_t^{\text{MR}}} \sum_{j \in \mathcal{J}_t^{\text{MR}}} \mathbb{1}_{j_i=j} \times \log \left(P_{ij}^{\text{MR}}(\tilde{\theta}^{\text{MR}}) \right)$$

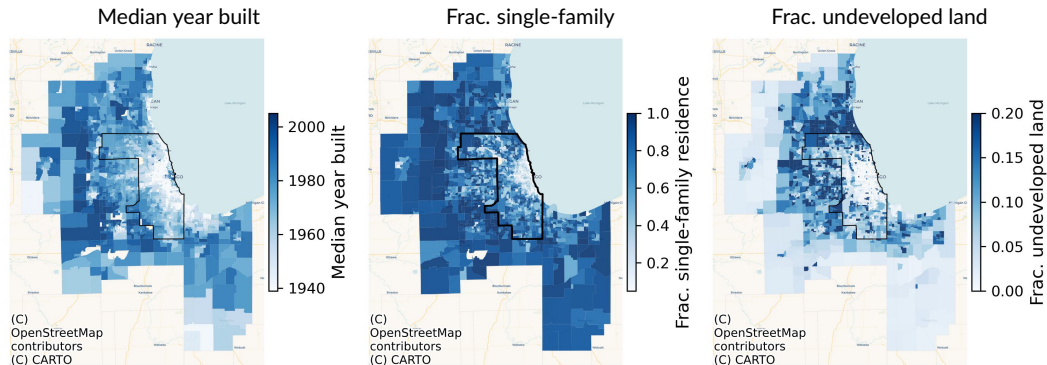
Use a contraction mapping to recover the mean utilities ($\tilde{\delta}$):

$$\tilde{\delta}_{jt}^{\text{new}} = \tilde{\delta}_{jt}^{\text{old}} - \log \left(\frac{\sum_{i \in \mathcal{I}_t} P_{ij}^{\text{MR}}(\tilde{\theta}^{\text{MR}})}{s_{jt}} \right)$$

Regress mean utilities on characteristics:

$$\hat{\delta}_{jt} = \gamma_0 \mathbf{x}_{jt} - \beta_0 r_{jt} + \xi_{jt}$$

Spatial correlation in common instruments



Note: This figure maps neighborhood characteristics that are commonly used to form instruments for rent. Median year built and fraction single-family residences are sourced from the 2019 5-year ACS. The fraction of undeveloped land is sourced from the National Land Cover Database (NLCD).

Index each unique combination of the (discrete) individual characteristics with b . Utility is:

$$\begin{aligned} v_{ij} &= \theta_b^{\text{beds}} * \text{bedrooms}_j + \theta_b^{\text{PUMA}} * \text{PUMA}_j + \theta_b^{\text{building}} * \text{buildingType}_j + \varepsilon_{ij} \\ &= v_{b(i)j} + \varepsilon_{ij} \end{aligned}$$

We parameterize each θ_b as: $\theta_b = \theta_{b,\text{industry}} + \theta_{b,\text{age}}^{\text{beds}} + \theta_{b,\text{married}} + \theta_{b,\text{kids}}$

The probability an individual with characteristics b selects option j is then given by:

$$P_{jb} = \frac{\exp(v_{bj})}{\sum_{j' \in \mathcal{J}} \exp(v_{bj'})}$$

Define N_b as the baseline number of individuals with characteristics v and g_{bt} as the growth relative to the baseline. Construct instrument as follows, then take Z-score:

$$z_{jt} = \frac{\sum_b g_{bt} N_b \hat{P}_{jb}}{\sum_b N_b \hat{P}_{jb}}$$

Only would decline if receive multiple offers. We assume:

- i) Offers arrive in random order, and a household accepts the first offer it receives
- ii) Households are not strategic with respect to offer/acceptance probabilities when choosing whether to apply to each affordable housing option

To compute acceptance probability:

- Accepting depends on number of other offers received conditional on offer at j
- Probabilities of applying and receiving an offer at each j are iid Bernoulli
- By Le Cam's, distribution of offers is approx. Poisson w/ rate $\rho_{ij} = \sum_{j' \in \mathcal{J}^{AH}, j \neq j'} P_{ij'}^{\text{Apply}} \times P_{ij'}^{\text{Offer}}$
- Conditional probability is then

$$P_{ij}[\text{accepts} \mid \text{offered}] \approx \sum_{n=0}^{\infty} \underbrace{\left(\frac{e^{-\rho_{ij}} \rho_{ij}^n}{n!} \right)}_{\text{Prob of } n \text{ other offers}} \underbrace{\left(\frac{1}{1+n} \right)}_{\text{Prob accept if } n \text{ other offers}}$$

Move-out rates

Population	Market-rate		LIHTC-eligible		LIHTC	
	1-year	3-year	1-year	3-year	1-year	3-year
Aggregate	0.7325	0.3411	0.7314	0.3664	0.7747	0.4779
White (non-Hispanic)	0.7249	0.3167	0.7273	0.3453	0.7592	0.4331
Black (non-Hispanic)	0.7604	0.3803	0.7524	0.3785	0.7734	0.4753
Hispanic	0.7611	0.3989	0.7658	0.4210	0.7789	0.4939
Any children	0.7383	0.3626	0.7301	0.3720	0.8091	0.4974
Joint filers	0.7407	0.3432	0.7712	0.4070	0.8491	0.5569
Household income: non-filer	0.7238	0.4019	0.7184	0.4059	0.8800	0.5325
Household income: (\$0, \$10k]	0.7161	0.3215	0.7163	0.3316	0.7181	0.4169
Household income: (\$10k, \$20k]	0.7240	0.3280	0.7309	0.3484	0.7509	0.4467
Household income: (\$20k, \$30k]	0.7337	0.3414	0.7472	0.3766	0.7998	0.4921
Household income: (\$30k, \$40k]	0.7412	0.3407	0.7529	0.3800	0.8243	0.5192
Household income: >\$40k	0.7395	0.3375	0.7372	0.3760	0.8416	0.5267

Note: This table documents the share still living in a unit 1- and 3-years after moving in.

Compute model-predicted average $\mathbf{w}$ across AH households each period

$$\hat{m}_t^{(q)} = \frac{\sum_{i \in \mathcal{I}_t} \left(w_i \times \sum_{j \in \mathcal{J}_t^{\text{AH}}} P_{ijt}^{\text{alloc}} \right)}{\sum_{i \in \mathcal{I}_t} \sum_{j \in \mathcal{J}_t^{\text{AH}}} P_{ijt}^{\text{alloc}}}$$

where P_{ijt}^{alloc} is the equilibrium probability that household i is allocated to affordable housing option j in period t .

For P_{ijt}^{accept} , approximate distribution of # other offers a household receives conditional on offer at j as a Poisson with arrival rate $\rho_{ijt} = \sum_{j' \in \mathcal{J}_t^{\text{AH}}, j' \neq j} (P_{ij't}^{\text{apply}} \times \pi_{j'k(i)t})$ (Le Cam, 1960), such that

$$P_{ijt}^{\text{accept}} \approx \sum_{n=0}^{|\mathcal{J}_t^{\text{AH}}|-1} \underbrace{\left(\frac{e^{-\rho_{ijt}} \rho_{ijt}^n}{n!} \right)}_{\text{Prob of } n \text{ other offers}} \underbrace{\left(\frac{1}{1+n} \right)}_{\text{Prob accept if } n \text{ other offers}}$$

For households endowed with an AH unit $j_i^0 \in \mathcal{J}^{\text{AH}}$, their move out probability is

$$P_{it}^{\text{moveout}} = \frac{\sum_{j \in \mathcal{J}_t^{\text{MR}}} \exp(\delta_{jt} + \lambda_{ijt})}{\exp(\delta_{j_i^0 t} + \lambda_{ij_i^0 t}) + \sum_{j \in \mathcal{J}_t^{\text{MR}}} \exp(\delta_{jt} + \lambda_{ijt})}$$

For moments, we compute the mean probability of moving out each period, as well as the covariance with household characteristics:

$$\text{Means:} \quad \hat{m}_t^{(q)} = \frac{1}{|\mathcal{I}_t|} \sum_{i \in \mathcal{I}_t} P_{it}^{\text{moveout}}$$

$$\text{Covariances:} \quad \hat{m}_t^{(q)} = \frac{1}{|\mathcal{I}_t| - 1} \sum_{i \in \mathcal{I}_t} (\mathbf{w}_i - \bar{\mathbf{w}}_i) \left(P_{it}^{\text{moveout}} - \bar{P}_{it}^{\text{moveout}} \right)$$

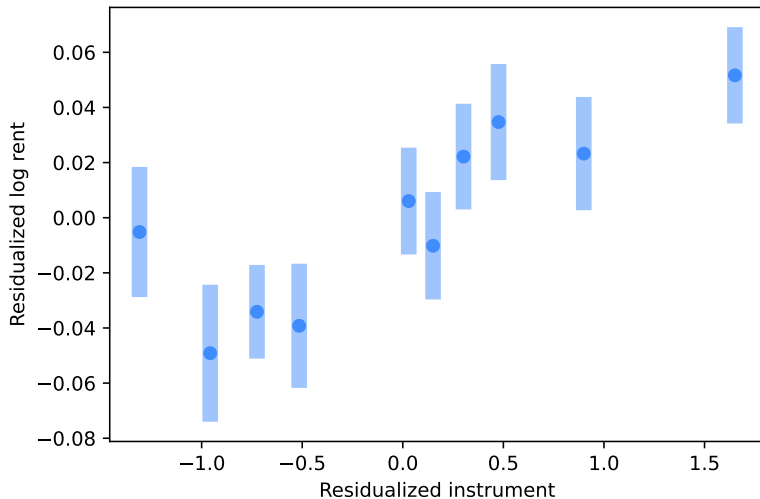
We use annual move-out rates to match the construction of the ACS

Instrument: potentially endogenous amenities

Outcome	Outcome mean	Coef. on instrument
# restaurants	2.051	0.008562 (0.0044)
# personal services	1.178	0.009641 (0.005)
# grocery stores	1.008	-0.01620 (0.0038)
# general retail	0.4508	-0.01064 (0.0048)
# entertainment	0.2946	-0.002269 (0.0035)
Frac. White (non-Hispanic)	0.5607	0.003697 (0.0017)
Frac. Hispanic	0.1971	-0.002002 (0.0015)
Frac Black (non-Hispanic)	0.1665	-0.002546 (0.0008)
Frac with college degree	0.4250	0.005678 (0.0033)
Frac housing vacant	0.0891	-0.001284 (0.0008)
Log median household income	11.16	0.01193 (0.0052)
Log population density	8.487	0.0004986 (0.0033)

Note: Establishment counts are based on the Business Register. Other neighborhood characteristics are based on the ACS. The first column is the mean across Chicago MSA neighborhoods. The second column regresses the outcome on the z-score of our instrument for market-rate rents with fixed effects for neighborhood and year. Standard errors are presented in parentheses.

Variation in instrument & rents

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Note: This figure documents the relationship between the instrument and rents, controlling for neighborhood fixed effects.

Covariate	Coefficient on rent
Exposure instrument	0.3099 (0.0524)
2 bedrooms	2.736 (0.0973)
3+ bedrooms	5.762 (0.1346)
Building <25yo	2.323 (0.095)
Big apartment building	0.3375 (0.1425)
Small apartment building	-1.543 (0.0883)

Note: Standard errors are presented in parentheses.

Value of LIHTC unit compared to similar market-rate

Population	Neighborhood opportunity idx		
	p25	p50	p75
All eligible	-0.9965 (0.0238)	-0.6746 (0.0162)	-0.3528 (0.0133)
High-need	-0.6647 (0.0331)	-0.2911 (0.021)	0.0826 (0.0174)
Moderate-need	-1.286 (0.0327)	-1.009 (0.023)	-0.7323 (0.0201)
Black or Hispanic	-0.0248 (0.0265)	0.1348 (0.0161)	0.2943 (0.0135)
White or other	-2.342 (0.0344)	-1.795 (0.0206)	-1.248 (0.019)
High-need, Black/Hispanic	-2.214 (0.059)	-1.541 (0.0323)	-0.869 (0.0275)
Moderate-need, Black/Hispanic	-2.405 (0.0424)	-1.921 (0.0266)	-1.437 (0.0255)
High-need, White/other	-0.005 (0.0373)	0.2415 (0.0221)	0.488 (0.0172)
Moderate-need, White/other	-0.0502 (0.0399)	-0.0025 (0.0258)	0.0453 (0.0215)

Note: These table documents the disutility of affordable housing in dollars (\$00s) relative to an observable-equivalent market-rate unit. Bootstrapped standard errors are in parentheses.

Offer probability comparison

Population	Neighborhood opportunity idx		
	p25	p50	p75
High-need	0.2365 (0.0057)	0.2236 (0.0103)	0.2098 (0.0153)
Moderate-need	0.1682 (0.0049)	0.1794 (0.0089)	0.1915 (0.0133)
Black or Hispanic	0.2235 (0.0036)	0.2225 (0.0035)	0.2205 (0.0034)
White or other	0.1675 (0.0049)	0.1689 (0.0048)	0.1716 (0.0047)
High-need, Black/Hispanic	0.1982 (0.0085)	0.1865 (0.0142)	0.1745 (0.0199)
Moderate-need, Black/Hispanic	0.1523 (0.0054)	0.1602 (0.0061)	0.1701 (0.0084)
High-need, White/other	0.2528 (0.0065)	0.2394 (0.0095)	0.2248 (0.0138)
Moderate-need, White/other	0.1858 (0.0081)	0.2006 (0.0146)	0.215 (0.0205)

Note: These table documents the probability of getting an offer in a world where all households apply to a new development with equal probability, set such that there are 5 applicants per unit. Bootstrapped standard errors are in parentheses.

Subsidies awarded through competitive allocation process

1. State specifies scoring criteria (e.g., points for nbhd chars.)
2. Developers submit applications
3. Credits awarded to highest-scoring applications

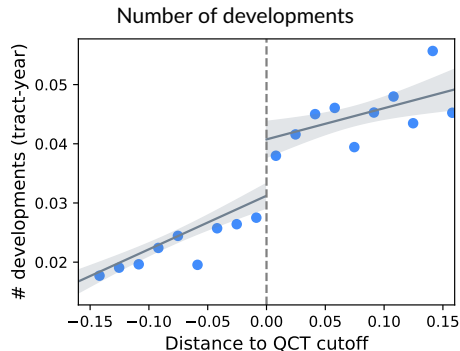
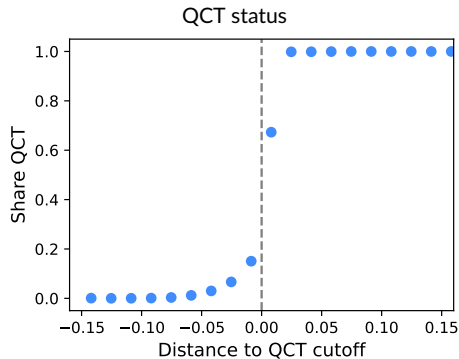
Size of subsidy depends on 'qualified basis'

- Baseline: all non-land construction costs and a development fee (usually 15%)
 - Spatial variation: +30% if in Qualified Census Tract (QCT) or Difficult to Develop Area (DDA)
 - QCTs defined using poverty rate and share of households eligible
 - DDAs defined using ratio of LIHTC to market-rate rent
- Useful variation at threshold

LIHTC supply-side: developments around QCT thresholds

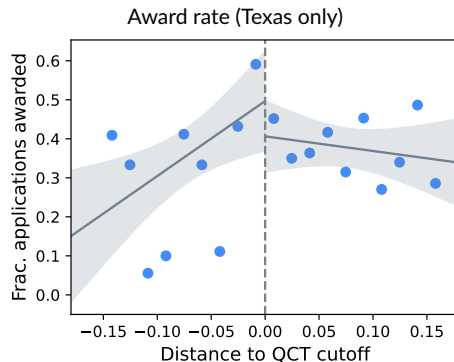
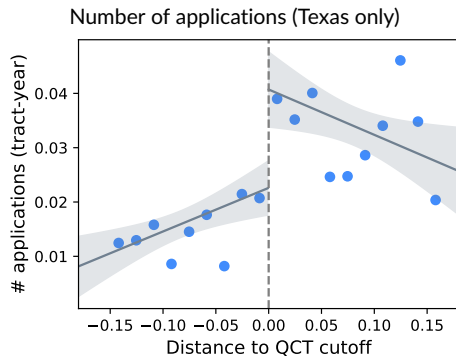
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LIHTC development around QCT threshold



Note: These figures document the distance to the threshold HUD uses to define Qualified Census Tracts (QCTs). Developments built in QCTs receive a 'basis boost' or 30%. The sample covers developments built in the 100 most populous metro areas between 2000 and 2015. Gray shading represents the 95% confidence interval.

LIHTC development around QCT threshold



Note: These figures document the distance to the threshold HUD uses to define Qualified Census Tracts (QCTs). Data cover LIHTC applications for credits in Texas between 2000 and 2015. Gray shading represents the 95% confidence interval.

Future LIHTC households vs. other eligible households

	Chicago MSA		50 Sample MSAs	
	(1)	(2)	(1)	(2)
Share non-Hispanic Black (2010)	0.1293 (0.0247)	-0.0264 (0.0230)	0.0679 (0.003)	-0.0008 (0.0027)
Share Hispanic (2010)	-0.0602 (0.0135)	0.0116 (0.0125)	0.0102 (0.0023)	0.0125 (0.0022)
Share w/ college (2010)	-0.0348 (0.0136)	-0.0130 (0.0138)	-0.0470 (0.0024)	-0.0183 (0.0024)
Overall opportunity index (percentile)	-0.0746 (0.0159)	-0.0142 (0.0141)	-0.0724 (0.0033)	-0.0199 (0.0031)
HUD jobs index	-1.241 (1.255)	0.1862 (1.227)	-0.7886 (0.2683)	0.6096 (0.2653)
Upward mobility index	-10.00 (2.032)	-0.9749 (2.026)	-6.681 (0.3161)	-1.420 (0.3017)
HUD school index	-5.330 (1.844)	-0.4356 (1.743)	-6.088 (0.3231)	-1.487 (0.3037)
Log med. hh income (2010)	-0.1010 (0.0304)	0.0015 (0.0291)	-0.1090 (0.0052)	-0.0407 (0.0050)
Household chars. used in model		✓		✓

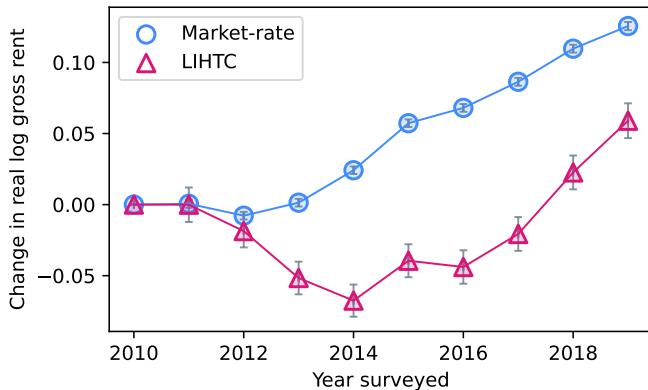
Note: This table presents a series of regressions of neighborhood characteristics on an indicator for whether the head of household moves into a LIHTC unit in the next two years. All specifications have CBSA interacted with year fixed effects. Standard errors are presented in parentheses.

Housing quality in American Housing Survey (AHS)

	Market-rate mean	LIHTC coefficient	Public housing coefficient
Has maintenance issue	0.2127	-0.0223 (0.0151)	0.0424 (0.0155)
Seen rodents last 3mo	0.1123	-0.0087 (0.0166)	0.0523 (0.0213)
Seen roaches last 3mo	0.1536	-0.0298 (0.0179)	0.0506 (0.0173)
Has barred windows	0.1837	-0.0883 (0.0335)	-0.0739 (0.0228)
Unit square feet	1323	-378.4 (64.23)	-373.9 (91.42)
Year built	1960	42.9 (0.6901)	9.447 (0.7544)

Note: Sample includes 2013 and 2015 American Housing Survey, subset to units in the 50 sample CBSAs. The coefficients are from a regression of a housing characteristic on indicators for whether the unit is LIHTC or public housing (with market-rate being the holdout group) and fixed effects for neighborhood (PUMA) and year. Market-rate means are weighted by the number of LIHTC units in the PUMA. Standard errors are presented in parentheses.

Rent growth: LIHTC and market-rate



Note: This figure documents the growth in real rents between 2010 and 2019 in the American Community Survey, split by LIHTC and market-rate. Each point is a coefficient from a regression of log gross rent on indicators for the year of the survey. 95% confidence intervals are represented by gray bars.

Goal: estimate the 'implicit subsidy,' or the opportunity cost of using a unit for LIHTC instead of market-rate

Method: compare LIHTC rent to estimate of market-rate rent for same unit

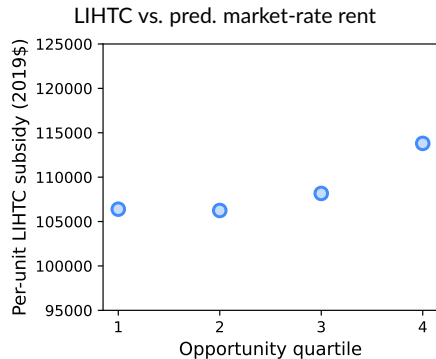
- Hedonic regression to predict rent for each LIHTC unit as if it were a market-rate unit
 - **Training data:** ACS market-rate units (2016-2018, Chicago MSA)
 - **Characteristics:** PUMA*year fixed effects, bedroom fixed effects, rooms per bedroom, year built bins, building type fixed effects (SFR, small apartment, big apartment)
- Key assumption: difference in unobserved characteristics between LIHTC and market-rate is constant across neighborhoods

Cost of new development #2: tax credit amount

Goal: estimate # tax credits that would be awarded for given neighborhood

Method: data on past allocations + XGBoost to predict awards based on neighborhood and development chars. ($R^2 = 0.54$ on holdout)

- **Development chars.:** # units, whether fully affordable, target population (if any), for vs. non-profit, target AML, and indicators for receiving other sources of funding (e.g., a state bond).
- **Neighborhood chars.:** predicted land price, WRLURI (with an indicator if missing), housing vacancy rate, log population density, log median household income, fraction White, fraction with college, fraction below poverty, log number of offices within 1 mile, log number of parks within 1 mile, and whether the tract is a Qualified Census Tract or Difficult Development Area.



Recent Supreme Court case on LIHTC siting and segregation

2008 Texas sued over LIHTC siting by a community organization

- *“...perpetuates racial segregation”* because of its *“failure to correct the disproportionate allocation of housing tax credits to low income minority areas”*

2012 District court agrees; mandates Texas prioritize ‘opportunity’ in future siting criteria

2015 On appeal, Supreme Court upholds validity of ‘disparate impact’ argument, but sets high bar of proof

2016 District court reverses decision; insufficient causal evidence

“CHIEF JUSTICE ROBERTS: Take two proposals. One is a proposal to build new housing in a low income area, it would benefit primary minorities; new housing, good thing. The other proposal is to build housing in a more affluent area. It would help promote integration of housing; also a good thing. Which one gets credit [...]? The one that is revitalizing a low income area or the one that is integrating a high income area?”

Use estimates from the Opportunity Atlas (Chetty et al., 2026) to approximate the causal effect on lifetime earnings of children, following steps from (Bergman et al., 2023)

1. Compute change in upward mobility rank between two neighborhoods
2. Multiply by 62% to approximate the *causal* effect of earnings rank at age 26 if moved at birth
3. Convert ranks into an estimate of lifetime individual earnings using ACS to compute lifetime income trajectory
4. Multiply by average number of children per LIHTC household in each quartile
5. Compute per unit-month value as present discounted effect (3% discounting) on earnings, divided by 18×12 to convert to monthly

Final estimate: +\$229 per unit-month for units in Q1, +\$450 per unit-month in Q4

Use estimates from (Diamond and McQuade, 2019) to estimate effects on neighboring renters, homeowners, and landlords

- Map their eight classifications of Census block groups onto our measure of neighborhood opportunity
 - They split by above/below 50% Black/Hispanic residents + quartiles of median income (in distribution of block groups with LIHTC development)
 - Key assumption: constant effects within each of their neighborhood classifications
- Use their per-household estimates for each group, inflated to 2019\$ and multiplied by number of renters, homeowners, and landlords within 1.5 miles
- Divide by 82 to get per-unit number, then amortize over 15 years at a 3\$ discount rate

Final estimate: –\$248 per unit-month in Q1, –\$451 per unit-month in Q4

Single city with options, $j \in \mathcal{J}^{\text{AH}} \cup \mathcal{J}^{\text{MR}}$, each with rents r_j and chars. $\mathbf{x}_j$

- Options are tuples of:
 - Neighborhood:** Public Use Microdata Areas (PUMAs)
 - Housing type:** (# bedrooms) $\times$ (single-family, small apt. building, big apt. building) $\times$ (<25 yo)
- $\mathbf{x}_j$ include nbhd demographics & amenities, # of bedrooms, building type, and age ▶

Single city with options, $j \in \mathcal{J}^{\text{AH}} \cup \mathcal{J}^{\text{MR}}$, each with rents r_j and chars. $\mathbf{x}_j$

Households endowed with characteristics $\mathbf{w}_i$ and current housing j_i^0

- $\mathbf{w}_i$ includes measures of current and past income, race/ethnicity, household size, family structure, age of household head, ...

Single city with options, $j \in \mathcal{J}^{\text{AH}} \cup \mathcal{J}^{\text{MR}}$, each with rents r_j and chars. $\mathbf{x}_j$

Households endowed with characteristics $\mathbf{w}_i$ and current housing j_i^0

Household i 's indirect utility for j is

$$u_{ij} = \underbrace{\gamma_i' \mathbf{x}_j}_{\text{Housing \& nbhd chars}} + \underbrace{\beta_i r_j}_{\text{Rent}} + \underbrace{\xi_j}_{\text{Unobserved quality}} + \underbrace{\varepsilon_{ij}}_{\text{T1EV}}$$

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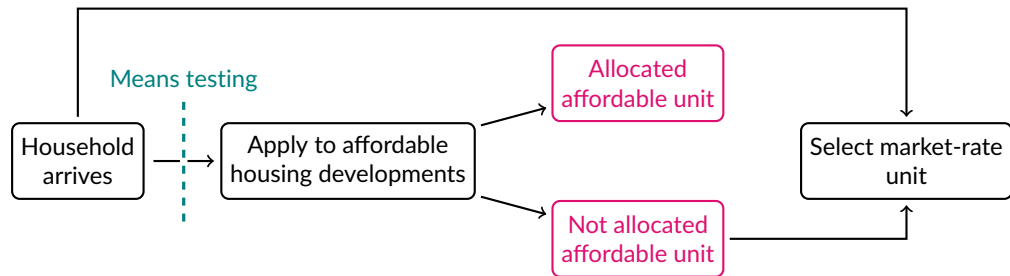
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- Preferences for AH (α_{ig}) also vary by opportunity index: $\alpha_{ig} = \alpha_0 + \sum_{\ell} (\alpha_{0\ell} + \alpha_{1\ell} \text{opp}_g) w_{i\ell}$
 $\rightsquigarrow$ 'Gap' between AH and observably-equivalent market-rate unit can vary by nbhd opportunity

Residential choice with affordable housing options



Static model, with two key ingredients:

1. Preferences for housing
2. Rationing process for affordable housing

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Assumption 1 (applications). Households can apply to each AH development without cost

$\implies$ Apply to $j \in \mathcal{J}^{\text{AH}}$ if better than endowed & market-rate options:

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$$\begin{aligned} P_{ij}^{\text{apply}} &= \underbrace{\mathbb{1}_{j \in \mathcal{J}_i^{\text{AH}}}}_{\text{Eligibility indicator}} \times \Pr \left[u_{ij} > u_{ij'} \quad \forall j' \in \{j_i^0\} \cup \mathcal{J}^{\text{MR}} \right] \\ &= \mathbb{1}_{j \in \mathcal{J}_i^{\text{AH}}} \times \left(\frac{\exp(\delta_j + \lambda_{ij})}{\exp(\delta_j + \lambda_{ij}) + \sum_{j' \in \{j_i^0\} \cup \mathcal{J}^{\text{MR}}} \exp(\delta_{j'} + \lambda_{ij'})} \right) \end{aligned}$$

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2) Developers allocate among applicants

Developers observe characteristics $\mathbf{w}$ for each applicant

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Assumption 2 (mechanism) Applicant i to development j receives an offer with probability

$$P_{ij}^{\text{offer}} = \frac{\exp(\phi_{ig(j)} - \zeta_j)}{1 + \exp(\phi_{ig(j)} - \zeta_j)}$$

where

- $\phi_{ig(j)} = \phi'_{g(j)} \mathbf{w}_i$ shifts applicant i 's relative offer probability, with loadings $\phi_g = \phi_0 + \phi_1 \text{opp}_g$ that depend on neighborhood opportunity index
- ζ_j is a stringency term set in equilibrium to clear the market

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E.g., waitlist with screening on $\mathbf{w}$, with stricter screening if demand is higher

1) Households apply to each development

2) Developers allocate among applicants

3) Offer probabilities clear the market

In equilibrium, stringency term ζ_j adjusts to clear the market

$$\underbrace{s_j}_{\text{Supply}} = \sum_{i \in \mathcal{I}} \underbrace{P_{ij}^{\text{apply}} \times P_{ij}^{\text{offer}}(\zeta_j) \times P_{ij}^{\text{accept}}}_{\text{Probability } i \text{ is allocated } j}$$

Estimating demand for a rationed good is challenging without applications

- Observed quantities $\neq$ demand at observed price
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Data: repeated cross-sections of market-rate choices from ACS

Method: follows Bayer, Ferreira and McMillan (2007), with new instrument for rent

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Intuition: Prefs. from step 1 inform who would move in/out, up to α and ϕ

Step 1: Estimating preferences using market-rate choices

Estimation similar to Bayer, Ferreira and McMillan (2007)

[▶ Details](#)

1. Aggregate to two-year periods, indexed by t
2. MLE to estimate heterog. pref. parameters; contraction map to get mean utilities (δ)
3. Regress mean utilities on characteristics:

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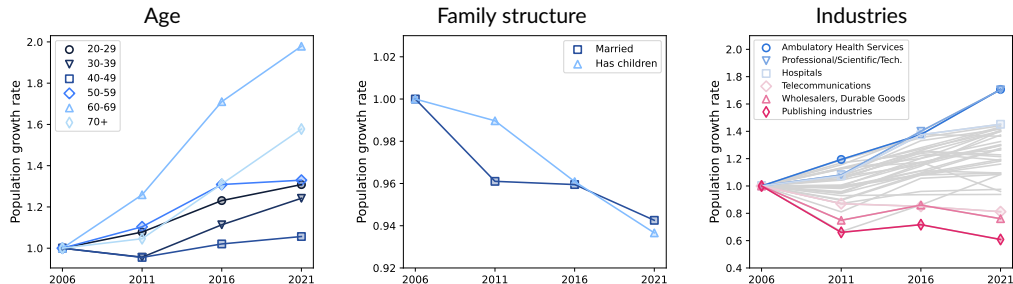
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- **Standard approach:** use characteristics of nearby neighborhoods ($\sim$ BLP instruments)
 - Potential issues if unobservables spatially correlated [► Spatial corr.](#)
- **Our approach:** isolate shifts in residual supply due to broad trends in demographics
 - Similar in spirit to Waldfoegel instruments (Waldfoegel, 2003; Berry and Haile, 2016)

Step 1: Instrument for rent

[◀ Back](#)

Large changes in the demographic and industry composition of cities over time



Note: Data are 5-year Public Use Microdata Samples (PUMS) from the ACS and include all individuals ≥ 20 yo living in one of the sample MSAs. Industry is 3-digit NAICS.

Large changes in the demographic and industry composition of cities over time

Instrument leverages how exposed a housing option is to these shifts

- **Intuition:** options popular among growing groups → increasing rent over time
 - Housing supply is inelastic (Saiz, 2010; Baum-Snow and Han, 2024)

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- Implement as a shift-share [▶ Details](#)
 - **Shares:** Using pre-period choices to estimate share of individuals (including homeowners) of each type choosing each option
 - Individual types based on (age bin, family structure, industry)
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- Add neighborhood fixed effects to isolate within-neighborhood differences

$$\hat{\delta}_{jt} = \gamma_0 \mathbf{x}_{jt} - \beta_0 r_{jt} + \psi_{g(j)} + \xi_{jt}$$

Estimated parameters: OLS vs. IV

[◀ Back](#)

	OLS
Gross rent (00s)	-0.0217 (0.0033)
2 bedrooms	0.2858 (0.0272)
3+ bedrooms	0.4452 (0.0329)
Building <25yo	0.4468 (0.0229)
Big apartment building	0.6445 (0.0276)
Small apartment building	0.5129 (0.0262)
Neighborhood fixed effects	✓
F-statistic	
N	5300

Note: The dependent variable is the mean utility of each housing option in each period. Neighborhood characteristics do not vary over time, so are absorbed into the fixed effects. For the BLP-style instruments we use four characteristics in a 3-6 mile ring around the focal neighborhood: the level of development and the fraction of land that is forested in the 2011 National Land Cover Database, the fraction of land defined as 'flat plains' from the US Geological Survey's National Elevation Database, and the share of units that are single-family in the 2010 5-year ACS. The sample size is rounded per Census disclosure requirements. Standard errors are clustered at the PUMA level and are reported in parentheses.

[▶ Instrument variation](#)[▶ Nbhd chars](#)[▶ First stage](#)

Estimated parameters: OLS vs. IV

[◀ Back](#)

	OLS	IV
Gross rent (00s)	-0.0217 (0.0033)	-0.1673 (0.0494)
2 bedrooms	0.2858 (0.0272)	0.6844 (0.1381)
3+ bedrooms	0.4452 (0.0329)	1.253 (0.277)
Building <25yo	0.4468 (0.0229)	0.7725 (0.1145)
Big apartment building	0.6445 (0.0276)	0.6939 (0.0403)
Small apartment building	0.5129 (0.0262)	0.2889 (0.0805)
Neighborhood fixed effects	✓	✓
F-statistic		35.03
N	5300	5300

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[▶ Instrument variation](#)[▶ Nbhd chars](#)[▶ First stage](#)

Estimated parameters: OLS vs. IV

[◀ Back](#)

	OLS	IV	BLP-style IV
Gross rent (00s)	-0.0217 (0.0033)	-0.1673 (0.0494)	-0.1784 (0.0586)
2 bedrooms	0.2858 (0.0272)	0.6844 (0.1381)	0.7113 (0.1653)
3+ bedrooms	0.4452 (0.0329)	1.253 (0.277)	1.310 (0.3276)
Building <25yo	0.4468 (0.0229)	0.7725 (0.1145)	0.7991 (0.1325)
Big apartment building	0.6445 (0.0276)	0.6939 (0.0403)	0.7059 (0.0394)
Small apartment building	0.5129 (0.0262)	0.2889 (0.0805)	0.2728 (0.0961)
Neighborhood fixed effects	✓	✓	
F-statistic		35.03	30.52
N	5300	5300	5300

Note: The dependent variable is the mean utility of each housing option in each period. Neighborhood characteristics do not vary over time, so are absorbed into the fixed effects. For the BLP-style instruments we use four characteristics in a 3-6 mile ring around the focal neighborhood: the level of development and the fraction of land that is forested in the 2011 National Land Cover Database, the fraction of land defined as 'flat plains' from the US Geological Survey's National Elevation Database, and the share of units that are single-family in the 2010 5-year ACS. The sample size is rounded per Census disclosure requirements. Standard errors are clustered at the PUMA level and are reported in parentheses.

[▶ Instrument variation](#)
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Step 2: Estimating AH utility (α) and lottery weights (ϕ)

[◀ Back](#)

Step 1 estimates $\rightarrow$ who *would* live in AH, up to α and ϕ

α : which *households* prefer AH, relative to similar market-rate unit

ϕ : which types of households are favored in screening process

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[◀ Back](#)

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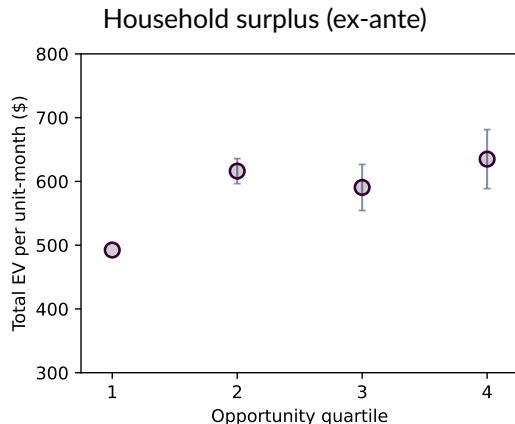
ϕ : which types of households are favored in screening process

Use GMM to match moments based on moves in and out of AH

- **Intuition:** screening parameters (ϕ) affects move-ins, household preferences (α) affect both
- **Move-in moments** ▶
 - Average of each household characteristic in $\mathbf{w}$
 - Covariance between household characteristics and nbhd opportunity index
- **Move-out moments** ▶
 - Average move-out rates from AH
 - Covariance between moving out and household characteristics
 - Covariance between moving out and household characteristics $\times$ nbhd opportunity index

[▶ Move-out rates](#)

Household surplus increases in neighborhood opportunity



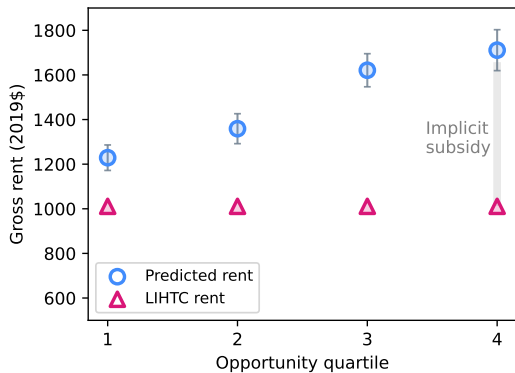
Note: Gray bars represent 95% bootstrapped confidence intervals for the *difference* from Q1 surplus. The standard error for Q1 is \$185.

Measure as equivalent variation (EV)

- A household's EV for development depends on:
 1. Probability allocated
 2. Utility if allocated vs. if not allocated
- Convert to rent-equivalent units (using β_i)
- Sum across all renter households

Costs increase more steeply than household surplus

Implicit subsidy of LIHTC

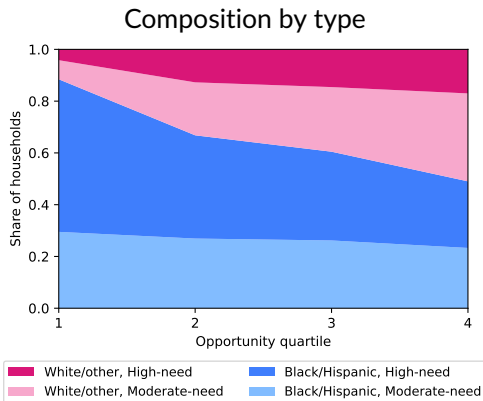


Note: LIHTC rent is based on the 60% AMI income limits, averaged across distribution of bedroom sizes. Gray bars represent 95% confidence intervals.

Measure costs as the 'implicit subsidy'

- Gap between unit's LIHTC rent and prediction of its market-rate rent
 - ~ Opportunity cost of renting as LIHTC
- 18% discount in Q1, 41% discount in Q4

Who benefits from a new development?

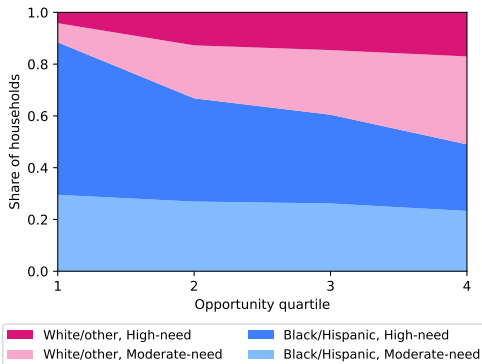


Note: Application (allocation) probabilities are for applying (being allocated) to any unit in the simulated development. Household surplus is computed as the equivalent variation in monthly rent based on differences in expected utilities pre/post-new development. 'High-need' refers to households whose predicted future income is in the bottom quartile of the nationwide distribution of renters, adjusted for household size and age. Gray bars represent 95% confidence intervals from bootstrapped standard errors.

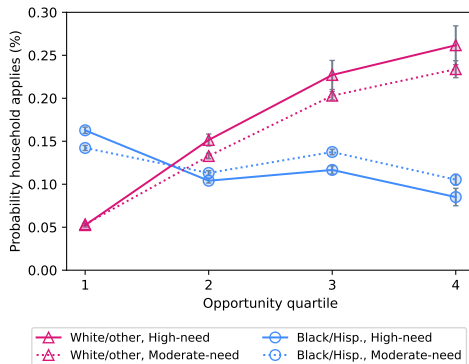
Who benefits from a new development?

[◀ Back](#)

Composition by type



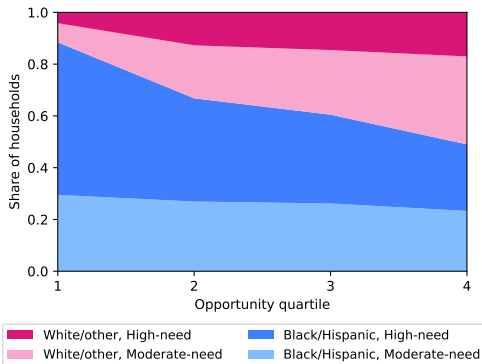
Probability *apply*



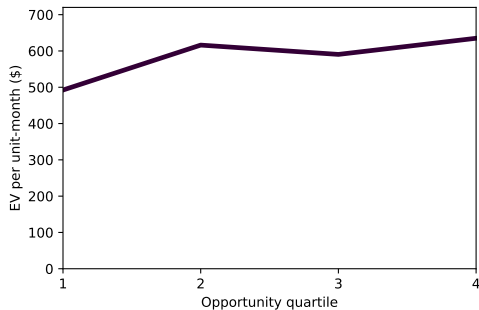
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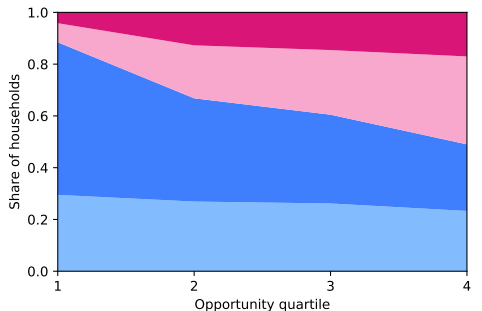
Household surplus



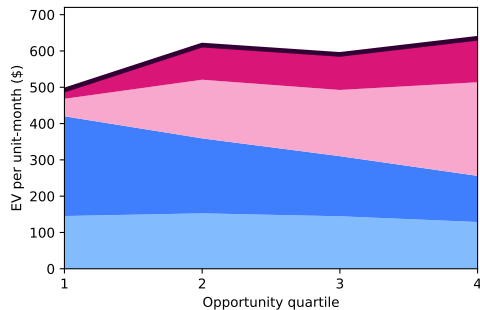
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Who benefits from a new development?

Composition by type



Household surplus



Note: Application (allocation) probabilities are for applying (being allocated) to any unit in the simulated development. Household surplus is computed as the equivalent variation in monthly rent based on differences in expected utilities pre/post-new development. 'High-need' refers to households whose predicted future income is in the bottom quartile of the nationwide distribution of renters, adjusted for household size and age. Gray bars represent 95% confidence intervals from bootstrapped standard errors.

Long history of court cases on whether AH perpetuates segregation

- 2015 Supreme Court case specifically on effects of LIHTC ▶
 - Increased policy focus on building in less segregated nbhds (Owens and Smith, 2023)

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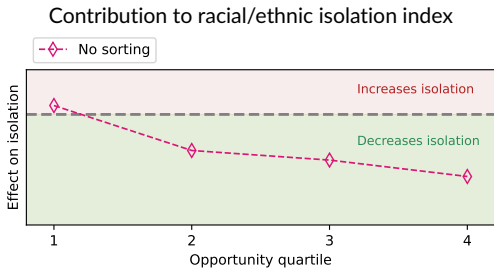
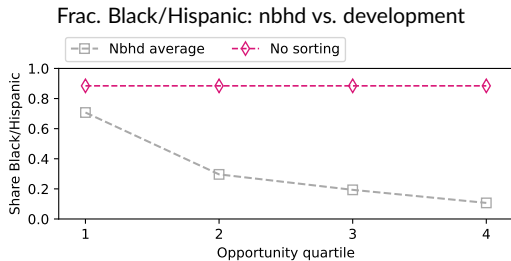
- 2015 Supreme Court case specifically on effects of LIHTC [▶](#)
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Measure marginal effect of siting on racial/ethnic and economic segregation

- Isolation index defined for groups A and B as:

$$\text{Isolation} = \underbrace{\frac{1}{|A|} \sum_{i \in A} \text{frac}A_{g(i)}}_{\text{Avg. exposure to A by A}} - \underbrace{\frac{1}{|B|} \sum_{i \in B} \text{frac}A_{g(i)}}_{\text{Avg exposure to A by B}}$$

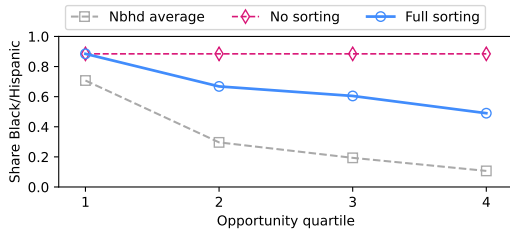
Racial/ethnic integration



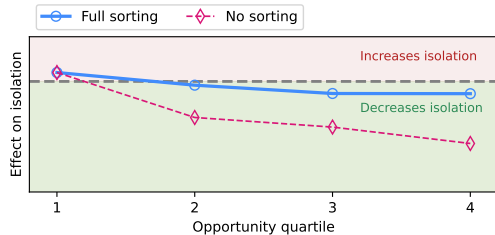
Economic integration

Racial/ethnic integration

Frac. Black/Hispanic: nbhd vs. development



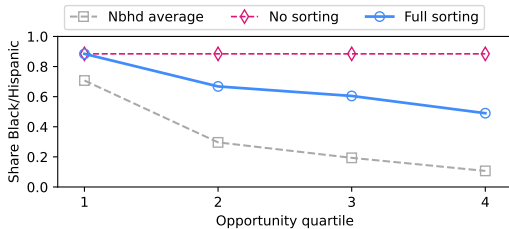
Contribution to racial/ethnic isolation index



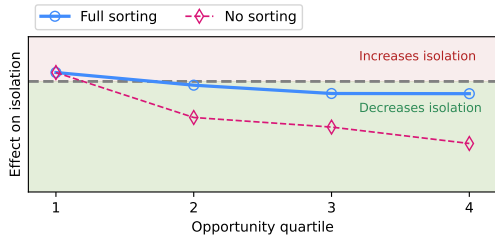
Economic integration

Racial/ethnic integration

Frac. Black/Hispanic: nbhd vs. development

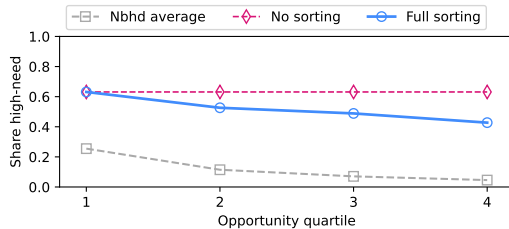


Contribution to racial/ethnic isolation index



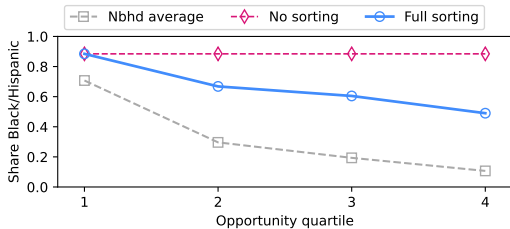
Economic integration

Frac. high-need: nbhd vs. development

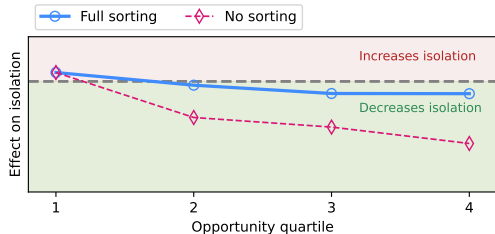


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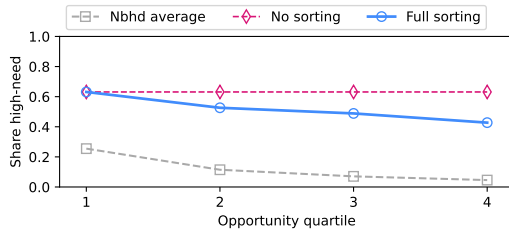


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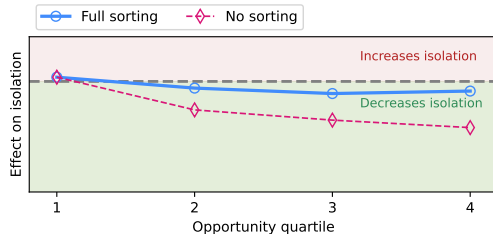


Economic integration

Frac. high-need: nbhd vs. development



Contribution to economic isolation index



Effects on children

[▶ Details](#)

- [Chetty et al. \(2026\)](#): Large causal effects of moving to 'high-opportunity' neighborhoods
- For LIHTC, moderated by fewer kids & changes to where households move from
- Approx. effect: **+\$220 per unit-month** for Q1 vs. Q4 development
 - +\$143,400 per child (birth to age 18)

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Effects on neighbors

[▶ Details](#)

- [Diamond and McQuade \(2019\)](#): LIHTC has positive spillovers in low-income, high-minority neighborhoods and *negative* spillovers in high-income neighborhoods
- Approx. effect: **-\$203 per unit-month** for Q1 vs. Q4 (15-year amortization with 3% discounting)
 - -\$46,000 per unit

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~> Removes any influence of private developers

Effects of post-construction policy changes

[◀ Back](#)

Policy	Share high-need		Share Black/Hispanic	
	Q1	Q1 → Q4	Q1	Q1 → Q4
Baseline	0.6312 (0.0244)	-0.2038 (0.0270)	0.8844 (0.0234)	-0.3943 (0.0302)

Note: This figure documents the effects of counterfactual processes or parameters on a range of outcomes for developments built in the bottom quartile (Q1) versus top quartile of neighborhood opportunity (Q4). 'Income-based rents' charges households 30% of their income at the time of application. 'Local preferences' requires that at least 50% of new tenants come from the surrounding neighborhood. 'Fair lottery' sets developer lottery weights to zero.

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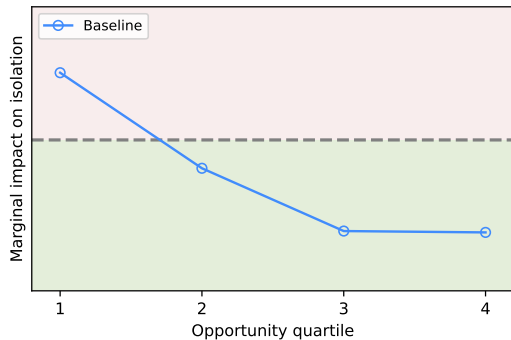
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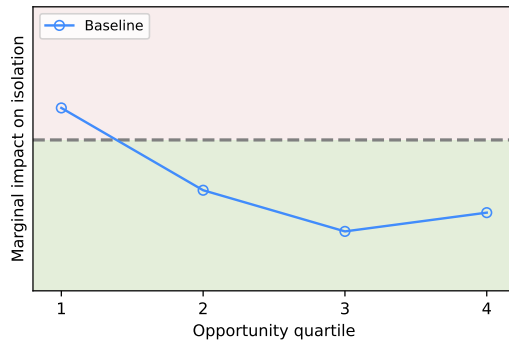
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Effects of counterfactuals on integration

Racial/ethnic isolation



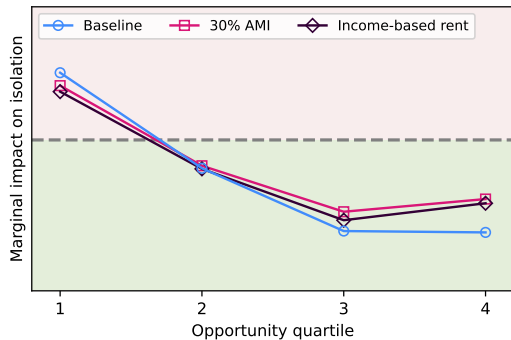
Economic isolation



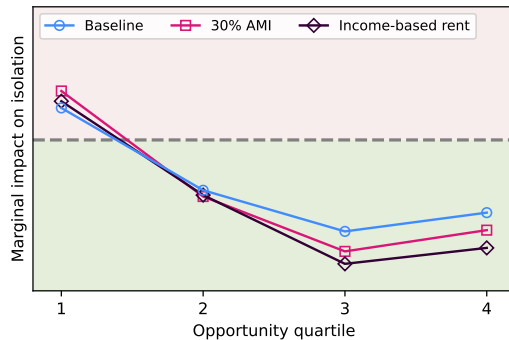
Note: This figure documents the effects of counterfactual processes or parameters on residential integration. 'Income-based rents' charges households 30% of their income at the time of application. 'Local preferences' requires that at least 50% of new tenants come from the surrounding neighborhood. 'Fair lottery' sets developer lottery weights to zero.

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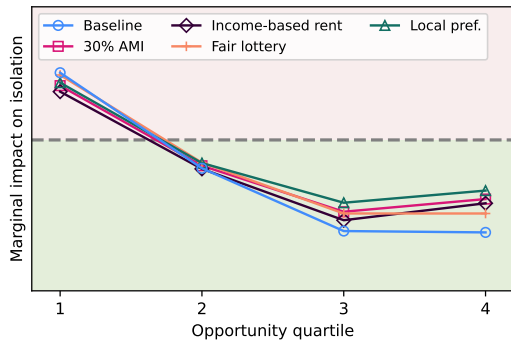
Economic isolation



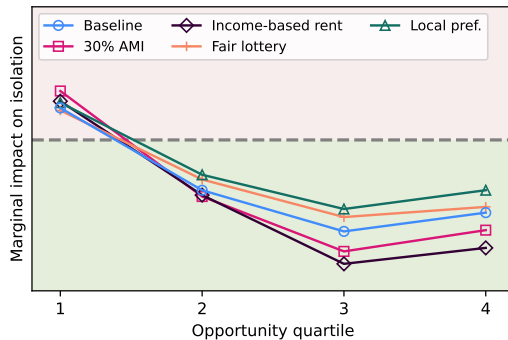
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